

EXHALE: Update Log (stage 3, from 2026-09-23)

Contents

1. The Ritz residuals of the preconditioned spectrum (2026-09-23)	2
2. Two records the campaign left open, closed (2026-09-23)	4
3. Two statements of the source that were not true, corrected (2026-09-23)	4
4. Four more source comments that were not true, corrected (2026-09-23)	5
5. One equivalent-width operator for the observation and every model (2026-09-23)	6
6. The stage export of the stationary alternation (2026-09-23)	6
7. Three defects the family C audit found, fixed, and two small ones (2026-09-23)	7
8. The regression check of section 7, a cell writer, and corrections from the spectral and On stall tests (2026-09-24)	8
9. Three defects the transient measurements found, repaired (2026-09-24)	10
10. The elemental flux records ride on the Riemann face mass flux; a row writer for the helium element (2026-09-24)	10
11. The molecular $K_{zz} = 1e9$ ladder certified across the observed equivalent width (2026-09-24)	12
12. The regression of sections 9 and 10, and the validation of the crossing (2026-09-24)	13
13. The width of the line on the molecular crossing, and the catalog summary brought up to date (2026-09-24)	14
14. The user's decisions of 2026-09-24 on the open items, and what each gave	16
15. The grid of the crossing (decision item 3) and the regression of sections 13 and 14 (2026-09-24)	18
16. The user's decisions of 2026-09-24 on the four remaining items	18
17. The atomic-data audit of 2026-09-24, applied	19
18. The H2-front energy imbalance, the H3+ non-LTE interpolant, and the descent below He/H 1.45 (2026-09-24)	20
19. The second round of the atomic-data audit (2026-09-24/25)	21
20. The molecular $K_{zz} = 1e9$ family under the audited physics (2026-09-25)	23
21. The code-size appendix moves to this log (2026-09-25)	24
22. The originals supplied on 2026-09-25, read and applied	24
23. The Markdown documents move to md/ (2026-09-25)	26
24. The last items of the MoCHII comparison, and the molecular crossing under the corrected physics (2026-09-25/26)	26
25. The helium-hydrogen charge exchange from its originals (2026-09-26)	28
26. A trial state no longer stops the run at an unclosed lower ghost (2026-09-26)	29
27. The molecular family under the corrected physics: catalog publication and the width of the line (2026-09-26)	30
28. The catalog summary brought to the tree's state, and why the molecular winds do not certify (2026-09-26)	31
29. Errors fixed and models solved first: the plan of 2026-09-26	32
29.1 An OpenMP data race in the metal sweep (fixed)	32

29.2 Fix batch (errors confirmed in the source)	32
29.3 The interdiffusion enthalpy flux (new physics, serious)	33
29.4 He+ + H non-radiative charge exchange near threshold	33
29.5 Golden's, fixture, suites	33
29.6 Models solved	34
29.7 A progress-control defect of the composition relaxation	34
29.8 Incidents of this series	35
29.9 The lower-ghost H2 closure stalled with a zero move (fixed)	35
29.10 The _adv post-process made second order	35
29.11 The lower face at the reservoir level	36
29.12 A restartable state after every outer pass	37
29.13 The unsolved molecular cases by continuation (state of 2026-09-28 16:30)	37
Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-09-22)	39
Fortran source inherited unchanged from ATES	48

This log continues md/Update_EXHALE_stage2.md from 2026-09-23. The earlier logs are kept under their own names:

- docs/Update_EXHALE_stage0.{tex,pdf}: the early phase (the ATES fork up to 2026-08-15).
- docs/Update_EXHALE_stage1.{md,tex,pdf}: sections 1-171, 2026-06 to 2026-09-05; "Update_EXHALE section N" in the source comments and the older memos points into that file.
- docs/Update_EXHALE_stage2.{md,tex,pdf}: sections 1-17, 2026-09-05 to 2026-09-22.

The entries of 2026-09-23 were first written into the stage 2 log as its sections 18 to 24 and moved here the same day as sections 1 to 7; a record of that day that cites "md/Update_EXHALE_stage2.md section 18" to "section 24" means section 1 to section 7 of this file (the references known at the move were rewritten).

Each entry below is one dated section: what changed, why, which files, the check that was run and its result. Section numbers restart at 1. The plan the entries execute is md/PLAN_20260923_rev2.md.

1. The Ritz residuals of the preconditioned spectrum (2026-09-23)

The spectrum hook EXHALE_PRECOND_SPECTRUM=1 printed Ritz values with no accuracy of their own (md/ISSUES_20260909.md item (D8), plan section 16). It now prints the residual $\|A y - \theta y\|$ of each.

The residual. For a Ritz pair $(\theta, y = V_k s)$ with $\|s\| = 1$ the Arnoldi relation $A V_k = V_k H_k + h(k+1,k) v(k+1) e_k^T$ gives $A y - \theta y = h(k+1,k) s_k v(k+1)$, so the residual is $|h(k+1,k)| |s_k|$. ritz_values_of_a_matrix_free_operator now hands back $h(k+1,k)$, and the routine prints that residual and the residual over the value's magnitude for the six smallest values of every compression. That form assumes a linear operator and an orthonormal basis, and the finite-difference action is linear only to its truncation, so for the three

smallest values of the whole operator the residual is also measured by applying the operator once more to y , once to each part of a complex pair. The operator action is now one routine, `action_of_the_preconditioned_operator`, called by the recursion and by the measured residual alike; the complex-pair packing of `dgeev` is undone in `coefficients_of_the_ritz_vector`. Every existing [diag 15] line is printed as before, and the residuals are new lines.

Verification. On the stored state of `atomic_scalar_gj1132x0.01_kzz1e9/HeH9.7 (g0002_20260919T004843Z_f7485b14)` with `EXHALE_TRUST_REGION=1`, `EXHALE_ELEM_DIAG=1`, `EXHALE_PRECOND_SPECTRUM=1` and `EXHALE_JFNK_MAXIT=1`, a 17-second run: the 30 [diag 15] lines of the binary before the change are byte-identical after it, so extracting the action moved nothing in the recursion; and the Arnoldi estimate and the measured residual agree to five or six digits for all three measured pairs, the complex one included. MEASURED. The hook runs only under `EXHALE_PRECOND_SPECTRUM=1` inside the trust-region step, which no case of the regression matrix enables, so the matrix was not rerun. Two unused local declarations of the same file (`islot`; `tv`, `net`) were removed on the way; they carried no value.

What it shows. At 200 products that spectrum is the one phase 9 of the campaign quoted (largest $9.99385\text{E-}01$, smallest $2.52128\text{E-}04$, ratio $3.96381\text{E+}03$, reproduced to every digit), and its small end is not resolved: $h(k+1, k) = 0.168501$, and the residual over the magnitude is 12.8, 3.58, 1.86, 1.86, 1.09 and 1.09 for the six smallest values, every one larger than one.

The whole space. `n_ritz_step` became `EXHALE_RITZ_STEPS=<k>` (default 200, capped at the number of unknowns, below 2 refused). At $k = 1500$, the number of unknowns of this operator, the Hessenberg is similar to the operator, and the same fixture took 136 s. MEASURED: smallest $5.60717\text{E-}06$, ratio $1.78560\text{E+}05$, 43 values below $1\text{E-}4$ and 319 below $1\text{E-}2$, against 0 and 6 at 200 products. So "nothing below $1\text{E-}4$ " is wrong on that checkpoint, and "there is no near-null mode of the preconditioned operator" is refuted. What the recursion finds are small eigenvalues of the operator the Krylov cycle sees: the hydrodynamic block alone (three unknowns a cell; `nvar_jac` = 3 + `nspec_row`, with no species rows on this route), scaled and preconditioned. [Corrected later on 2026-09-23 by `md/lhs1140b_spectral_checks_20260923.md`: this read "and shifted by the pseudo-time diagonal"; the shift is in the band preconditioner only, the hook being handed `idtau_leg` = 0.] They are not established as time scales of the atmosphere, and they say nothing about the alternated composition-hydrodynamics map. [Wording corrected 2026-09-23 after the review of `md/PLAN_20260923.md`: this paragraph read "it has a cluster of slow modes".] Below about $1\text{E-}5$ even the full recursion does not resolve them: the residual measured on the operator is $9.72\text{E-}06$, $9.88\text{E-}06$ and $9.56\text{E-}06$ for the three smallest values (the first two one conjugate pair), larger than the pair's magnitude and 0.96 of the third's, so none is resolved from zero at the production probe length; the floor was later measured to be the rounding of the forward difference (it falls as $\text{arc}^{-0.97}$ with the probe length, spectral checks memo). What the measurement bounds rigorously is the smallest singular value of the action as applied: on the normalized Ritz vector its norm is at most $|\text{thetal} + \text{llr}| = 1.5\text{E-}05$. Whether the $1\text{E-}5$ floor of the residual is truncation, rounding or nonadditivity of the finite-difference action is not decided by these runs; `measure_the_additivity_of_the_jacobian_action` is the narrower test. The vectors (the real and imaginary parts of the pair, and the third) carry 53 to 61 per cent in the mass rows and peak in the wind around cells 93 to 169, not at the base, but since they are not resolved their location is not established either. The documents that quoted the 200-product numbers carry a dated correction beside them. What stands is the

direct measurement they were offered beside: on the alternated route the Krylov cycle reaches its tolerance in one to four products.

2. Two records the campaign left open, closed (2026-09-23)

EXHALE_RESIDUAL now writes the configuration it measured under. The route stopped at `src/EXHALE_main.f90:1335`, before `write_setup_report` and `write_resolved_config` further down, so `EXHALE_setup.out` was left empty and `EXHALE_resolved.out` absent. It now writes both before its stop, as the direct steady route already did before its own. MEASURED on the stored state of `atomic_scalar_gj1132x0.01_kzz1e9/HeH9.7`: 62 and 48 lines written where there were none, the printed residual byte-identical (9.806145E-01, 8.723527E-06, 1.071147E+00), and the resolved file identical to a normal run's but for two added lines, the base ghost count and its electrons as solved, present because this route evaluates the residual, and so solves the ghost, before it writes.

The conversion now asserts the charge. `molecular_seed_from_atomic_state` computed the positive charge density before and after the $2\text{ H} \rightarrow \text{H}_2$ transfer and never compared the two. It now does, cell by cell, against `inventory_feasible_slack` (eight units in the last place) times the charge density, the bound `element_inventory` already uses for two expressions of the same nuclei; a neutral cell must come back exactly. A breach stops the run, and the largest change is printed in units of $\text{eps} * n_{\text{chg}}$. A relative change was not used because a neutral cell carries no charge and the ratio is not defined there. The `molecular_seed` suite asserts it in each of its three modes: 29 of 29 pass, and the change is 0.00 units in every mode, so in memory the transfer conserves the charge bitwise, and the 1.5 units the M2 conversion audit measured were its file round trip, as that audit said. The failure path was read, not exercised.

3. Two statements of the source that were not true, corrected (2026-09-23)

Found while checking the review of `md/PLAN_20260923.md` against the code; the plan's revision is `md/PLAN_20260923_rev1.md`. Neither change moves a number: one is a comment and one is the text of a printed line.

The certificate printed a withdrawn reason. Its report says why the species rows below `cert_regime_wind_r = 1.20` are reported and do not gate, and for the band 1.10 to 1.20 it printed that the row's discretization, $4.6\text{e-}4$ at the binding radius, "stands above any tolerance one could set there". The comment above `cert_regime_wind_r` said the same, that a $1\text{e-}5$ gate would ask the residual to fall below the error of its own discrete equation. That inference was withdrawn on 2026-09-11 (`md/code_status_20260910.md` section 3.3): the algebraic residual of the discrete equations is not bounded below by their truncation error, and the same routine took a wind H2 row to $2.4\text{e-}13$ on a held background. The gating decision is unchanged; its stated reason is now the one that document gives, that no state of this code has come near $1\text{e-}5$ there and that the terms of the row are conserved only to $2\text{e-}3$ (`certification.f90`, the comment of the two regimes and the report block).

The absent-carrier threshold claimed a round-off argument. The comment on `carrier_absent_fraction` (`diffusive_photochemistry.f90`) said that $1\text{e-}20$ of the free reservoir is "four decades below the round-off of a double against the reservoir, so nothing at or under it is a resolved density at all", and that such a carrier carries nothing

any reported quantity depends on. A small term vanishing from a rounded sum does not make it unresolved: stored as its own double, 4e-17 stays 4e-17 with a relative spacing of 1.5e-16, and its production, loss and transport stay resolved with it. The comment now says the threshold is a choice and not a round-off limit, and that the test reads the abundance alone, not the production or the incoming flux that could raise it. The threshold itself is unchanged.

Verified: the build compiles (EXHALE.x md5 69233669123c368b2d58aebc132eda85; the printed string moved it from 67646328ad476a976a7953aba65bdfc), and one stationary evaluate of a stored molecular state (the pass-6 state of kzz1e9/HeH0.55 from the 2026-09-21 alternation audit) prints the corrected block and reproduces to four digits the verdict that audit measured on it under its own binary (carrier balance H2 9.722E-04 at cell 500, elemental transport He/H 7.353E-04 at cell 336). The line counts of both files are unchanged. The regression matrix was not rerun: the printed text enters no output file the matrix compares, and a comment enters no object code.

4. Four more source comments that were not true, corrected (2026-09-23)

Found while checking the second review of the LHS 1140 b plan (md/PLAN_20260923_review1.md) against the code; the plan's revision is md/PLAN_20260923_rev2.md. All four are comments in diffusive_photochemistry.f90; the line count of the file is unchanged and the rebuilt EXHALE.x has the same md5, 69233669123c368b2d58aebc132eda85, so no object code moved.

- **The boundary section of the module header said "OUTER: zero gradient, as the element operator does",** and that a run whose carriers are not negligible there is flagged in output/Oxygen_chemistry.txt. The carrier operator imposes zero diffusive, eddy and settling flux on the outer face ($J_f(N) = 0$), which with a settling drift is not a zero gradient, and Oxygen_chemistry.txt is written only by oxygen-chemistry runs and flags no carrier. The header now states the flux condition and points to carrier_face_coefficients.
- **carrier_face_coefficients justified that closure by the outer cells being collisionless,** quoting $Kn_{bulk} = 1.4-2.4$ at the domain top and an exobase at 18.8-25.4 R_p . Those numbers come from runs made before the 2026-09-13 well-balanced re-solve. MEASURED with src/utls/collisional_validity.py on the current certified states and on the kzz1e9/HeH0.55 audit state: Kn_{bulk} at the top cell is 0.055-0.142, the exobase lies above the domain, and the largest species value in the critical region is 0.41-0.59. The comment now says the outer cells are transitional and that the zero face flux rests on the other reason it gives, the measured inward settling flux of a ghost-built face. md/collisional_validity.md carries the same measurement as a dated note.
- **The row-scale floor was said to be converged by construction** ("a density that small cannot change any observable, so a row below it IS converged", and "every term of such a row is round-off"). The floor is the rate that would move a carrier by 1e-20 of its element in one signal-crossing time, a scale that a row is judged against; it proves nothing about the row.
- **The absence test was said to separate round-off from a real row.** It reads the abundance alone, and a carrier far below the threshold can carry a real, resolved production; the comment now says that the test removes such a cell from the carrier gate without asking whether production or incoming flux could raise it, and that this is a choice.

Outside the source, the comment on the 2.26 km/s overlay shift in LHS1140b/make_memo_figures.py said the equiv-

alent width is unaffected. The equivalent widths are computed from unshifted arrays, so the shift enters none of them; applied to a model it would move the fixed-window EW by $1.2\text{e-}5$ relative (MEASURED). The comment now says both.

Verified: make rebuilt EXHALE.x to the same md5; the Python file parses. Nothing was run beyond that, since no executable statement changed.

5. One equivalent-width operator for the observation and every model (2026-09-23)

The user's decision on md/PLAN_20260923_rev2.md section 4: the observed statistic stays as it is ($1.1084293096 \pm 0.0294774727$ %A, working target 1.108 ± 0.030), and every model is put through the same operator. A model spectrum, already convolved to the instrument resolution, is interpolated linearly onto the observation's own vacuum samples inside 10832.60 to 10834.20 Å (44 samples, 10832.622 to 10834.175 Å) and integrated with their trapezoid weights. Before, each model was integrated with a masked trapezoid on its own grid inside the nominal window, so the two integrals started and stopped at different wavelengths.

One module, six readers. LHS1140b/he10830_equivalent_width.py holds the window, the vacuum/air ratio, the observation, and the operator (model_equivalent_width, transit_file_equivalent_width, observed_equivalent_width). The six places that computed an EW each on its own now call it: LHS1140b/models/status.py (the results table and, since this change, the EW column of the provenance table, which it used to copy from each run's REPRODUCE.md), LHS1140b/make_memo_figures.py (the unbroadened and the broadened curves, and the observed value, whose variable and printed label no longer call the released spectrum a digitization), LHS1140b/models/compare_archive.py, LHS1140b/models/write_reproduce.py (whose line also said the window was in air; it is in vacuum), and the inline EW of LHS1140b/models/run_case.sh. The unused regular expression that read the EW out of REPRODUCE.md was removed.

What moved, MEASURED. Over the 177 EXHALE spectra of the catalog the change is at most $1.45\text{e-}4$ %A, relative $-1.1\text{e-}3$ to $+0.9\text{e-}3$; 31 spectra move in the fourth decimal, 15 of the 83 certified rows of the catalog table among them (for example the certified molecular $K_{zz} = 1\text{e}9$ He/H 2.13 and 9.7 read 1.5604 and 2.4060 where they read 1.5603 and 2.4061), and one crossing moves ($K_{zz} = 0$: 3.1011 to 3.1010). Broadened curves and p-winds curves move more, up to about 0.6 per cent, because their wings reach the window edges that the old model integral kept and the observation's samples do not (for example the matched-kernel EW of the well-mixed He/H 0.4 solution, 1.067 to 1.065). LHS1140b/MODELS.md sections 7 and 8 were regenerated by models/status.py --write; docs/lhs1140b_catalog_state.tex carries the new EWs, a paragraph on the operator and the recomputed comparison of its section 4 (1.374869 to 1.356745 %A, 0.60 standard deviations), rebuilt to seven pages and checked; the memo docs/lhs1140b_exhale_vs_pwinds.tex and its figures are brought to the same operator in a separate pass (md/lhs1140b_ew_operator_memo_changes_20260923.md).

6. The stage export of the stationary alternation (2026-09-23)

EXHALE_STAGE_EXPORT=1 (default off) writes, for every pass of the alternation, the state the driver holds at four named stages: hydro_entry (before the hydrodynamic solve; from the second pass on, the state the previous composition

update handed over), hydro_return (the accepted solve after the ionization sweep that follows it), certification (the joint test, with the three residual rows), and composition_return (after the element and carrier updates and the rebuild of p and T at the fixed conserved state). Each goes to output/stage_state_p<pass>_<stage>.txt with every row, ghosts included: r, rho, v, p, T, heat, cool, the particle and electron densities, H I, H II, He I, He II, He III, He(2³S), H2, H2+, H3+, HeH+, the conserved variables, and at the certification the residual rows. At hydro_entry and at the certification the driver also asks conservation_budget for the next assembly under the stage's name (conservation_budget_request_next, new), so the energy row's terms of that state are exported without counting assemblies, which probes and trials also consume. It is the instrument md/PLAN_20260923_rev2.md P1.5 and P1.6 (b) call for (the four named-stage captures of review R4).

The writer calls nothing that sets a module variable. MEASURED on one pass from the certified molecular_scalar_gj1132_kzz1e9/HeH2.13 g0008 state: the data rows of Hydro_ioniz.txt and Ion_species.txt are bitwise the same with the export off (twice), with it on, and with the previous binary 69233669123c368b2d58aebc132eda85; the log differs only by the export lines. The two exported budget files carry their stage names. Binary EXHALE.x md5 b017fe42407c980068e112a8ca3d71d3, copied to LHS1140b/models/EXHALE_b017fe42.x with its manifest. The regression matrix was not rerun: every new call in the driver is guarded by the key, and the budget module gained a subroutine and a header line that are reached only on a request, so with the key off no executed statement of any route changed. The bitwise check above covers the stationary alternation; the matrix's cases, which march, were not run.

7. Three defects the family C audit found, fixed, and two small ones (2026-09-23)

The audit of the 13-decade H2 hole in molecular_photochem_gj1132_kzzprofile/HeH9 (md/lhs1140b_family_c_h2_slab_20260923.md, item P1.7 of md/PLAN_20260923_rev2.md) traced it to the molecular seed and found why no solve removed it. Each finding was checked in the source before it was changed.

The element relaxation threw away its own kept steps (binary_element_diffusion.f90, relax_element_composition). Its retry budget counts every rejected step of the relaxation, and when it ran out the routine returned element_relaxation_failed and restored the entry composition, even after 24 to 31 steps had been kept. The module's outcome table, and the driver that reads it, reserve that outcome for a relaxation that kept no step; the driver then refuses the pass and skips the carrier relaxation (EXHALE_main.f90, the outer_element_update_refused branch). This is how the carrier operator never ran on the family C case in four solves, and how the 2026-09-18 rung at He/H 1.411 ended at its second pass ("took 28 steps and found no admissible 29th", READ, L34c section 4). A relaxation that has kept steps now ends as a partial advance (element_relaxation_step_budget) with the kept steps handed back, and says so once in the log; one that kept none is still refused with its entry composition. The retry budget still counts every rejection. Test: two new rows of src/tests/element_operator/ on a test knob (element_refuse_after_accepted_for_test, -1 in every production run) that lets three steps through and refuses the rest: the relaxation reports the partial advance, three steps, and a moved composition; with no step let through it reports the refusal and returns the entry composition bitwise. Every row of the suite passes.

A sweep could rewrite a transported H2 (ionization_equilibrium.f90, the extraction of the molecular partition).

With the carriers transported the sweep is handed H2 (`x_h2_fixed`) and solves the rest against it; a root meets the imposed value through its own row, but a non-root acceptance (class 4) or a clamp onto the element budget (class 3) returned another `x(4)`, and the extraction wrote that back as the cell's H2. MEASURED on the seed's intermediate state of the family C case: the returned H2 differed from the handed one at exactly the 21 non-root cells, by up to 82 per cent; with the fix it differs nowhere by more than $7.8\text{e-}16$. The imposed value is now written and the atomic hydrogen takes the remainder of the budget.

The molecular seed read a root off cells the sweep had not solved (`molecular_seed_from_atomic_state.f90`, the local mode). The seed takes the root of each cell's own H2 row at the ions of one sweep; at cells where that sweep adopted a non-root with every ion at zero, the root was $4\text{e-}17$ of the hydrogen, thirteen decades below the neighbouring cells, and that was the hole. The sweep now records the acceptance class of each cell (`ieq_cell_acceptance_class`, written and never read inside the sweep), and the seed reads no root off a class-4 or class-6 cell: such a cell takes the root of its nearest solved neighbours, interpolated in $\log x_2$ over r , and the smaller of that and the fit. MEASURED on the same reproducer: the 21 cells go from $4\text{e-}17$ (and from up to $6.8\text{e-}4$ where the rising slab was) to a smooth $1.8\text{e-}4$ to $2.1\text{e-}3$ between the solved cells 428 and 450; outside them the seed moves by at most $8.2\text{e-}4$ relative, through the H2 fix above. Two further changes the audit proposed were tried and not kept: iterating sweep and root to a common fixed point did not converge in 30 iterations and drove the base layer to its unshielded branch ($x_2 = 6.9\text{e-}4$ at 974 K), because a cell's shielding column is the H2 above it and the iteration is not a local one; and adopting the root in place of the smaller of root and fit follows from the same iteration and was withdrawn with it. The `molecular_seed` suite passes 29 of 29.

Smaller. The comment on `nH_free` in `diffusive_photochemistry.f90` said it excludes the hydrogen of the ion stages and molecular ions; it is the cell's hydrogen less the HeH^+ nucleus (`carrier_element_totals`). And `carrier_row_terms.txt` was written with ES14.7, which drops the E of a three-digit exponent ($1.0332735\text{e-}100$); it is now ES15.7E3.

Not changed, reported. Two provenance attributions in the state index of the family C case are wrong (the audit memo, D4): the certificate of g0002 belongs to a later solve, and g0001's manifest names another binary than its own file header. The index is written by the catalog tools and was not edited by hand.

Verified: the element-operator and molecular-seed suites pass; the H2 fix and the seed on the family C reproducers as above; the stationary route with `EXHALE_STAGE_EXPORT` unchanged. `EXHALE.x md5 45190de9ba3752eb720a728d54d3a1e3`. The one regression case that runs the H2 carrier operator, `mol_carrier`, was run with the binary before and after the element and H2 fixes (section 8).

8. The regression check of section 7, a cell writer, and corrections from the spectral and On stall tests (2026-09-24)

The one regression case that runs the H2 carrier operator. `mol_carrier` was run from its input with `OMP_NUM_THREADS=1 EXHALE_MAXSTEPS=12000` twice, once with the binary before the fixes of section 7 (`EXHALE_b017fe42.x`) and once after (`EXHALE_af853fe7.x`), in copies (`LHS1140b/models/.P1/d2d3_mol_carrier/`). MEASURED: the two `Hydro_ioniz.txt` differ by at most

1.3e-11 relative and the two `Ion_species.txt` by at most 9.4e-13; neither run met a non-root cell or the retry budget of the element relaxation, so the difference is the H2 now written exactly at the imposed value where a root had met it to the solver's tolerance. Below 0.1 per cent, it needs no golden refresh.

A cell writer on the EXHALE_NEWTON_TEST route (`EXHALE_main.f90`). The route printed only the row maxima of the solver's vector $F(Y)$, so where it differed from the `EXHALE_RESIDUAL` route the cell was not known (ISSUES D5). It now writes `output/newton_residual_profile.txt` with the columns of `output/residual_profile.txt` (its temperature column is 0: the route forms no temperature array). MEASURED on `molecular_scalar_gj1132_kzz1e9/HeH2.13_g0008` with `EXHALE_RELOAD_EQ=0`: the two routes differ in cells 1 and 2 alone, where the base face mass flux differs by 9 per cent, and nowhere else in 500 cells; the mass-row maxima (5.10e-09 against 1.37e-09) are that base difference.

From the checks of the preconditioned spectrum (`md/lhs1140b_spectral_checks_20260923.md`):

- the cost line of the additivity measurement printed the run's cumulative count of inner cell solves where it meant this measurement's own (`nt_calls`, now `nt_calls_here`, `steady_newton.f90`);
- the comment that the Newton model's products reuse the frozen WENO3 weights is true on the line-search route and not on the trust-region route, where the merit evaluation sets `weno_mode = 0` and nothing restores it before the leg and the spectral hook; the comment now says so, and which of the two the leg should use is left open;
- the base boundary's comment that the states "sit decades from" the thresholds of the wind window now records a state that sits one decade away and a probe that crosses the switch (`base_boundary.f90`);
- the records that called the spectrum's operator "shifted by the pseudo-time diagonal" (ISSUES D8, stage 3 section 1, the `p2_base` note, the plan rev2 section 3.6) were corrected: the shift is in the band preconditioner only.

From the On stall test (`md/lhs1140b_on_stall_end_to_end_20260923.md`):

- Ionization transport: True was refused with Coupled carrier solve: True but not with On stall, which hands the same registry the same unknowns once the alternation stalls; it is now refused with either (`input_read.f90`);
- the comment on `EXHALE_JFNK_MAXIT` said a named cap makes the run take one stationary solve; it forbids a second entry into the stationary route and caps the solve of every pass inside the one entry, which the test measured (`EXHALE_main.f90`); the plan rev2 carried the same error and is corrected. The handover itself fails: the conditions are met and the transition begins, and the re-evaluation of the snapshot's residual stops the run on the face mass flux guard, the carrier balance also reading a background the snapshot does not hold. That is a change of the solver's control flow and is reported for a decision, not fixed.

The update log. The entries of 2026-09-23 were moved from the stage 2 log (its sections 18 to 24) into this file on the user's instruction, and `src/utils/update_log_to_tex.py` now takes the stage as its argument (2 or 3; the code-size appendix stays with stage 2). `docs/Update_EXHALE_stage2.tex` was regenerated for sections 1 to 17 (310 pages, no error or undefined reference) and this log's twin built (7 pages at the time).

Verified: the element-operator and molecular-seed suites pass; the builds compile. `EXHALE.x md5 46d5b9f4c0a62c1ab0cdf9bb165839ef`.

9. Three defects the transient measurements found, repaired (2026-09-24)

The P4 measurements of the plan (md/lhs1140b_p4_transients_20260923.md, whose worker reported them and left them in place) found three defects; each is repaired and its path checked.

The default CFL number was a single-precision literal. `real*8 :: CFL = 0.6` stored 0.6000000238 (parameters.f90); every run that took the default marched at a step $3.97\text{e-}8$ longer than the one it declared (MEASURED: the first step on a certified molecular state is 0.54736376538957865 s with the default and 0.54736374363929763 s with CFL: 0.6). Now 0.6d0. The same kind of literal set the Voronov (1997) collisional-ionization coefficients of all seventeen rows of `Cool_coeff.f90` (11.26 stored as 11.2600002), and the physics probe `voronov_carbon_ionization.f90` mirrored it on purpose to compare exactly; both are now double-precision literals, and the probe passes at its $1\text{e-}12$ tolerance (MEASURED). The rounding was below $1\text{e-}7$ relative, four decades under the precision of the fits; a declared constant is still the constant it declares.

EXHALE_UPDATE_MAP could stop before its last requested step. The diagnostic switched off the flux-spread stop, the residual gate and the JFNK hand-off, but not the step-change stop or the stall stop, and on a stationary state a step of a quarter of the CFL step already moves the state by less than `dtu_th`: a six-step map ended after three blocks. While the map runs, every stop flag is now cleared after the stop tests of each step (`EXHALE_main.f90`), so the map ends on its own step count and the staged secondary ionization cannot change the physics between two of its steps; the stop line says "the update map wrote its N requested step(s)". MEASURED on a copy of the failing run without `Force_start`: six blocks, dt from $6.6296\text{e-}05$ to $6.6296\text{e-}11$ of the code unit.

EXHALE_DUMP_IC=1 wrote placeholders as the lower ghost rows. `load_IC` fills the primitive ghost rows with cell 1's state until the first `Apply_BC`, which writes the ghosts of the conserved array only; the dump wrote the primitive array. It now takes the ghost rows of the primitive array the boundary derivation held (`bc_W_hold`). MEASURED: the two lower ghost rows read $\rho = 1.302\text{e}14$ and $1.242\text{e}14$, $T = 233.0998$ and 226.0275 K, against the placeholder $4.857\text{e}13$ and 546.06 K before, and the file's 233.1004 and 226.0281 K (the $2.5\text{e-}6$ is the ghost composition the boundary re-derives from the loaded column).

The P4 memo now carries its command record (section 7, which its worker had not written) and the repairs beside each defect.

10. The elemental flux records ride on the Riemann face mass flux; a row writer for the helium element (2026-09-24)

What was wrong. The element transport operator is handed the face mass flux of the Riemann solve (`face_mass_flux_of_state`), the one the mass row differences. Two records of it were not: the base budget entry (`element_base_flux`) and `output/element_flux_profile.txt` built their face mass flux as the face mean of the cell-centred density and velocity. Near the base the cell-centred velocity carries a collocated two-cell odd-even mode (project record P44), so on the certified molecular_scalar_gj1132_kzz1e9/HeH0.55 state the file read $4\pi r^2 F = -1.07\text{e}9, +1.16\text{e}8, -3.8\text{e}7$ and $+5.9\text{e}7$ g/s at faces 1 to 4, while the Riemann flux is $3.934813\text{e}7$ g/s through every face

(MEASURED). The "standing base sound wave" the comments named was that mode, not a wave of the solution.

What the records are now (binary_element_diffusion.f90, certification.f90, EXHALE_main.f90):

- write_element_flux_profile takes the face mass flux of the state it describes and is called by the certification of that state, with the flux the element row was measured on, whenever EXHALE_DIFFUSION_CHECK=1 or a lower-atmosphere profile is in use; the element step no longer writes it. Before, the file held the profile of the last relaxation, taken before the last hydrodynamic solve, and a stationary evaluate never wrote it.
- The base budget entry uses the flux of its own evaluation: the relaxation's Frho_in, or on the marching path the step flux $F_0/6 + F_1/6 + 2 F_2/3$ of the three SSP-RK3 stages (the flux whose divergence is the step's density change), read once per step; with neither, it says "not measured".
- face_mass_flux_cgs converts a code face flux with $n_0 \mu v_0$ and the face mean of msum.
- The velocity is no longer an argument of element_diffusion_step or relax_element_composition (nothing read it).
- composition_residual and trace_composition_residual reduced their norm with max, whose result with a NaN is compiler dependent; a non-finite row now sets the norm to huge.
- element_transport_residual, documented as side-effect free, added its face reconstructions to the run's face counters; it now restores them.

The helium row, term by term. EXHALE_ELEMENT_ROW_TERMS=1 (default off) writes output/element_row_terms.txt from the certification's own evaluation: the diffusive and advective terms of each face, the residual, the floor, the scale and the measure, with the face F_{rho} , X_{face} and J . Its measure agrees with the certification's full-precision anchor to the sixteenth digit.

What it measured (md/lhs1140b_element_row_terms_20260924.md): on the certified He/H 0.55 and 1.8 states the helium flux through $r < 1.10$ varies by $2.2\text{e-}3$ and $7.8\text{e-}5$ of its median, nearly all of it the step across cell 2 (the first solved cell, row $6.1\text{e-}5$ and $2.3\text{e-}5$, ungated); every other cell below $r = 1.2$ stands at or below $1.5\text{e-}7$ and $2.7\text{e-}7$. At cell 2 the residual would change the helium density in $9\text{e}9$ and $3\text{e}10$ s, against an advective cell time of $2\text{e}6$ s and a diffusive one of 47 s. The comment in certification.f90 that gave "element fluxes not conserved, spread 9.8" as the reason for not gating $r < 1.10$ had measured the face-mean artifact; it and the printed line now say the band is reported by decision, and the gating itself is unchanged. The Phase-E statement that the LHS 1140 b overlap window is structurally empty rested on the same quantity and is marked stale where it stands (md/phase_e_flux_closure_design.md, src/utls/element_flux_closure.py, md/certification_tolerance_anchoring_20260910.md); it was not re-measured. src/utls/collisional_validity.py and md/collisional_validity.md used the same wording for the base cells; corrected (the computation, whose velocity scale binds four decades below its threshold there, is unchanged).

Checks. The solved state does not move: an evaluate of the certified He/H 1.45 state by the new binary writes Hydro_ioniz.txt, Ion_species.txt and both _adv files identical to the 46d5b9f4 evaluate apart from the provenance line, CERTIFIED, and its element_flux_profile.txt carries $7.0353671\text{e}7$ g/s from face 1 outward (MEASURED; the worker measured the same on the He/H 0.55 and 1.8 states). Suites: element operator 65 of 65 (its base record rows assert the new definition and fail against the old objects by 1.54 and 1.31), diffusion_tests.x 35 of 35,

certification 123 PASS, steady species rows 202, species face flux 18 (the last two by the worker). The dead velocity arrays of `diffusion_tests.f90` and `certification_contexts.f90` are removed, and the test T6, which claimed a uniform mixture "under an arbitrary velocity field" that the operator never read, now says what it tests.

11. The molecular $K_{zz} = 1e9$ ladder certified across the observed equivalent width (2026-09-24)

Items P1.1 to P1.3 of the plan, run under the binary `EXHALE_46d5b9f4.x` (the tree of section 8), every rung a stationary alternation from a certified parent mapped onto the new reservoir with the pressure held (`map_state_to_grid.py --ic --reservoir He/H`, the base mixing ratio from the same nucleus fraction f), 24 passes and a 3 h ceiling allowed. Run directories under `LHS1140b/models/.P1/`; driver `.P1/make_rung.sh`.

He/H	parent	passes	wall of the passes	outcome	EW [%A]	log10 Mdot
0.55	the 2026-09-21 audit state (pilot)	12, then 4	4.30 h, then 0.07 h	certified in the continuation	0.1015	7.59
1.8	certified 2.13 (descent pilot)	12, then 3	1.04 h, then 0.04 h	certified in the continuation	1.3584	7.88
1.45	certified 1.8	17	1.15 h	certified	1.0964	7.85
1.6	certified 1.8	21	1.22 h	certified	1.2155	7.86
1.52	certified 1.45	14	0.88 h	certified	1.1532	7.86
0.689	certified 0.55	24 (budget)	2.85 h	not certified: elemental row 9.9e-4, still falling	0.274 (uncertified state)	7.67

MEASURED. Every certified state was re-entered four times in a chain of `Restart intent: stationary evaluate`, each entry reading the state the one before wrote: **CERTIFIED** at every entry, the largest relative change of any field after four entries $1.2e-12$ (the 1.8 chain; $3.5e-13$ or less for the others). The EW is the catalog's measurement operator (section 5) on the spectrum `EXHALE_transit.py` synthesized from the published state. The two adjacent certified rungs 1.45 and 1.52 straddle the observed 1.108 ± 0.030 %A, and `models/status.py` puts the crossing of the log-log chord at $\text{He/H} = 1.4643$. The interval is 0.0205 dex, below the plan's 0.03 dex floor, so the refinement stopped there.

Connectivity (`.P1/connectivity/compare_rungs.py`, figure docs/figures/lhs1140b_kzz1e9_molecular_rungs.pdf): from 1.45 to 2.13 the temperature, velocity, density, H II and He 2^3S profiles, the mass-loss rate (7.847 to 7.902), the outer Mach number (0.515 to 0.508) and the H2 front ($x(\text{H}_2) = 1e-2$ at 1.200 to 1.161 R_p) move monotonically, and neighbor differences scale with the step (about 3 per cent in T for 0.02 dex). The 0.55 state is a different wind: $x(\text{H}_2)$ above 0.5 to 1.13 R_p and near $1e-2$ in the far wind, a peak temperature of 3400 K against 5200 K and more, He 2^3S thirty times lower. The interval between 0.55 and 1.45 is not sampled (the 0.689 rung is its first point), so whether the two are one branch, and whether a second crossing lies between them, is not established. The 0.689 rung's last state

(uncertified) is intermediate: $x(\text{H2})$ above 0.5 to 1.10 R_p as at 0.55, but $1.5\text{e-}4$ at 10 R_p , near the 1.45 value.

The line shape (docs/figures/lhs1140b_kzz1e9_molecular_ew.pdf): the crossing is of the integrated EW alone. At $\text{He/H} = 1.45$ the model red line is 4.07 per cent deep, 3.3 times the depth Cherubim et al. (2026) fit ($1.24 +0.22/-0.23$ per cent), and correspondingly narrower; the molecular solutions share the line-shape disagreement of the unbroadened atomic spectra.

Published (models/run_case.sh with SEED= the certified run, which solves nothing new from a certified entry, evaluates, synthesizes the spectrum and writes REPRODUCE.md): molecular_scalar_gj1132_kzz1e9/HeH0.55 g0007 and the new cases HeH1.45, HeH1.52, HeH1.6, HeH1.8 (g0002 each), added to models/make_models.py. The new cases' input.inp is the certified HeH2.13 input with the He/H changed, the input the states were solved under; make_models.py itself would write Restart intent: stationary equilibrate for a molecular case, which the older HeH0.083 carries and the certified cases do not (a difference of route, reported for a decision). The catalog inventory reads 157 state indexes, 100 certified. LHS1140b/MODELS.md sections 7 and 8 and models/CASE_INVENTORY.{md,json} were rewritten by status.py --write; docs/lhs1140b_catalog_state.tex carries the new rows (section 13) and docs/lhs1140b_exhale_vs_pwinds.tex its corrected molecular paragraph and campaign table (both rebuilt, no error, pages checked). A verification binary copied into models/ had become the "newest EXHALE.x" that status.py takes as the binary of record; it was moved to the run directory that used it.

The physical record of the straddling states (plan section 3.1, with the writer of section 10, .P1/element_row_terms/record_straddle.txt): the H2 carrier row is worst inside the gated wind ($5.16\text{e-}6$ and $7.90\text{e-}6$), and the helium element row is worst at cell 2, ungated ($2.04\text{e-}5$ and $1.27\text{e-}4$), the next worst below $r = 1.2$ being $1.07\text{e-}6$ (cell 6) and $5.4\text{e-}6$ (cell 3), both in the first cells above the base; at cell 2 the residual would change the helium density in $2.9\text{e}10$ and $4.6\text{e}9$ s, against an advective cell time of $1.5\text{e}6$ s, and the region below 1.2 passes the helium flux on to within $9.3\text{e-}5$ of its throughput.

P3 (md/lhs1140b_p3_low_xuv_20260923.md, run 2026-09-23 under EXHALE_69233669.x): the atomic He/H 9.7 branch certifies at 0.0675, 0.0662 and 0.065 of the fiducial XUV (EW 0.0239, 0.0172, 0.0132 %A) and refuses at 0.06 and 0.0624 on the elemental row; each certified rung re-entered four times, CERTIFIED throughout.

12. The regression of sections 9 and 10, and the validation of the crossing (2026-09-24)

The matrix (make check, binary ac232b34, the tree with section 9; sixteen cases run, against the goldens of 2026-09-20). Verdict REGRESSION FAIL, for three reasons, MEASURED and attributed with the binary of the day before (EXHALE_46d5b9f4.x, the tree without section 9) run on copies of the cases:

1. **The CFL literal of section 9.** The control reproduces the goldens (mol_lyman_werner identical, mol_carrier within $1.3\text{e-}11$), so every movement is today's, and in these marching cases the only change that acts is the step $3.97\text{e-}8$ shorter. The state files move by at most $9.1\text{e-}5$ (wasp_he23off; $1.6\text{e-}5$ or less in the rest), below the $1\text{e-}3$ tolerance. What exceeds it is the adv_mass_row column of Hydro_ioniz_adv.txt, the mass-row measure the post-processing writes per row, in the four molecular 12000-step snapshots that are far from stationary (du near 0.8): $1.2\text{e-}3$ to $4.3\text{e-}3$ relative (mol_lyman_werner, mol_diffusion, mol_sec_ion, mol_carrier), and 0 against $5.6\text{e-}15$ in one row of wasp_full_newton. hydrostatic_column moves one ghost-row ion density from

- 4.1e-185 to 1.2e-170 (the control is identical to its golden). None of these is a change of a physical quantity above the tolerance.
2. **oxygen_chemistry stops before its first step**, with the control as well: "ghost composition is not a fixed point", the map applied 20 times and still moving by 5.7e-7 against an accuracy of 1e-11, after the constrained molecular roots of the cold start were clamped onto the simplex. The golden (2026-09-20 13:53) predates the ghost fixed point (commit of 2026-09-20 16:18); the case has not run since. The refusal is the boundary's designed behavior, so what to do with it is reported for a decision, not changed.
 3. Two files of `iontrans_metals` have no reference (MISS, not a failure).

Section 10 does not move them. The five cases that exercise the changed paths (element operator and its records, the certification's element block) were re-run with the tree of section 10 (76c3ce17) against the outputs of the matrix: `mol_diffusion`, `lower_profile`, `iontrans_metals`, `mol_carrier`, `hydrostatic_column`, REGRESSION PASS, all byte-identical (tolerance 0). The other cases run without element diffusion and do not reach those paths.

The goldens were not refreshed. The refresh of the sixteen cases that section 9 moves was refused by the session's permission check; the outputs of the matrix stand in the case directories and are what a refresh would snapshot.

The validation of the crossing of section 11 (plan order 12):

- A reverse continuation: He/H 1.52 reached downward from the certified 1.6 certifies (pass 13) and agrees with the 1.52 reached upward from 1.45 to 5.3e-6 in every hydrodynamic field and 1.4e-4 in every species, EW 1.15317 against 1.15316 %A (LHS1140b/models/.P1/reverse_HeH1.52_from1.6/).
- A refined grid (LHS1140b/models/.P1/refine_grid/): the two straddling states on 1000 cells (the base block and every other spacing halved, the outer radius unchanged) certify (passes 16, 15) and give EW 1.0640 and 1.1196 %A (against 1.0964 and 1.1532 on 500 cells), log10 Mdot 7.84 and 7.85. They still straddle 1.108; the chord moves from He/H 1.4643 to 1.505. The discretization shift, 3 per cent in EW, is of the size of the observation's working error; a 2000-cell pair is running to read the order.
- The physical record of section 11 and the operator of section 5.
- The domain test waits for a grid option (a decision).

13. The width of the line on the molecular crossing, and the catalog summary brought up to date (2026-09-24)

The broadening the line needs (user instruction: after the EW, match the width with an assumed turbulent Gaussian as `docs/lhs1140b_exhale_vs_pwinds.tex` does). LHS1140b/models/.P1/connectivity/broaden_molecular.py restates the memo's operation (the excess absorption of the instrument-convolved spectrum convolved with a Gaussian of FWHM f in velocity, the three-Gaussian extractor of `he_line_metrics.py`, the EW operator of section 5, f matched where the red-pair FWHM reaches the measured 0.841 Å) and first reproduces the memo on its atomic well-mixed He/H 0.40 state: $f = 22.22$ km/s, depth 1.229, blue 0.161, ratio 7.65, EW 1.0645 (the memo prints 1.065). MEASURED on the certified molecular $K_{zz} = 1e9$ states:

He/H	cells	EW, $f = 0$	matched f [km/s]	red [%]	blue [%]	red/blue	EW at f
0.55	500	0.1015	22.32	0.113	0.014	7.89	0.0981
1.45	500	1.0964	22.21	1.224	0.160	7.64	1.0600
1.52	500	1.1532	22.20	1.287	0.169	7.63	1.1149
1.6	500	1.2155	22.19	1.357	0.178	7.61	1.1752
1.8	500	1.3584	22.16	1.516	0.200	7.58	1.3135
2.13	500	1.5604	22.12	1.741	0.231	7.55	1.5087
9.7	500	2.4060	21.80	2.684	0.359	7.47	2.3264
1.45	1000	1.0640	22.21	1.188	0.155	7.65	1.0288
1.52	1000	1.1196	22.19	1.250	0.164	7.64	1.0825
observed		1.108 +/- 0.030		1.24 +0.22/-0.23	0.25 +0.14/-0.12	6.7 +12.7/-3.1	1.108 +/- 0.030

At He/H 1.45 the width is reached at 22.21 km/s (sigma 9.43 km/s) and the depth, the blue component and the ratio then fall inside the observed 1 sigma intervals without being tuned. The kernel does not depend on composition or chemistry (22.12 to 22.32 over 0.55 to 2.13; the memo's atomic families 22.20 to 22.40). The Gaussian wings carry 3.3 per cent of the EW out of the window, so the broadened EW crosses the working target 1.108 at He/H 1.5112 between the certified 1.45 and 1.52 on the catalog grid; on 1000 cells the broadened EW of 1.52 is 1.0825, and the chord extended past 1.52 reaches the target at 1.55 (DERIVED, an extrapolation). On the advection-corrected profiles weighted by $n(2-3S) r$ over r , $c_s = 4.33$ km/s in the line-forming region of both straddling states, so the kernel is Mach 2.2 (sigma), 3.1 (sqrt 2 sigma, the v_{turb} of Lampon et al. and p-winds) and 3.8 (sqrt 3 sigma). Figure docs/figures/lhs1140b_kzz1e9_molecular_broadened.pdf.

A correction to the memo found on the way. The memo's c_s of the line-forming region of the well-mixed He/H 0.55 state, 4.1 km/s, did not follow from its own $T = 2499$ K and $\mu = 1.94$; recomputed with the memo's weighting ($n(2-3S) r$ integrated over r on the `_adv` profiles, which reproduces $T = 2498$ K and $\mu = 1.93$) it is 4.21 km/s, and the Mach numbers of the matched kernel move from 2.3, 3.3, 4.0 to 2.2, 3.2, 3.9 (summary and section of the memo corrected, marked in place).

The catalog summary. Renamed on the user's instruction to docs/lhs1140b_catalog_state.tex (no date in the name; it was lhs1140b_catalog_state_20260922.tex, and the references in md/du_stop_vs_stationary_20260920.md, md/session_handoff_20260920.md and the stage 2 and 3 logs now point to the new name; the stage 2 twin Update_EXHALE_stage2.{tex,pdf} still carries the old name and was not regenerated). Brought up to date rather than annotated: the counts (157 indexes, 100 certified; 103 models, 91 certified), the table by group, the table of certified models (eight new rows) and of uncertified ones (the molecular 0.55 row out, the He/H 9.7 rungs at 0.06 and 0.0624 in), the mechanisms (and the count of refusing rows, which the old text stated wrongly: the energy row, not the mass row, appeared most often), the operator-difference statement re-measured over the 91 certified rows (still at most 1.5×10^{-4} %A, 15 rows in the fourth decimal), and a new section 5 on the molecular crossing, its validation and the broadening. Rebuilt (11 pages, no error), every changed page checked.

14. The user's decisions of 2026-09-24 on the open items, and what each gave

The list of section 6 of `md/lhs1140b_plan_rev2_execution_20260924.md`, reordered by what it changes, was put to the user; every item was decided "as recommended". Items 2 and 3 (the rungs below He/H 1.45, and the grid of the catalog) are still running and are recorded in the next section.

Item 1, the width of the line. The added Gaussian of section 13 is an empirical parameter of the comparison, fixed by the measured width alone; every composition read from the full line is stated as conditional on it (`docs/lhs1140b_catalog_state.tex` section 5, the execution record).

Item 4, which cells gate the species rows: compared, not changed. A default-off diagnostic, `EXHALE_CERT_GATE_DIAGNOSTIC=1` (`certification.f90`), prints for every carrier and element row what the verdict would be if every cell from cell 3 outward were gated against the wind tolerance ($1e-5$), cells 1 and 2 reported. MEASURED on all 99 ladder cases re-evaluated with it (`md/lhs1140b_gating_diagnostic_20260924.md`): every stored verdict reproduces; under the candidate rule 38 of the 91 certified cases stay certified and 53 would be refused, 52 of them by the helium element row in the first cells above the base (cell 3 in 45 cases, up to $9.1e-2$) and one by the H2 row (molecular `kzz1e9/HeH9.7`, $4.76e-5$ at $r = 1.127$); no uncertified case would become certified. In 27 atomic `K_zz` cases the element row exceeds $1e-5$ in 76 to 205 cells up to $r = 1.09$ to 1.19 (on the atomic `kzz1e9 He/H 1.50` and `1.60` states $1.5e-3$ and $1.8e-3$ at cell 3, 184 and 182 cells out to $r = 1.16$). The premise of the item, that below $r = 1.2$ only the first solved cell stands above $1e-5$, holds on the six certified molecular `K_zz = 1e9` states and not on the atomic ladders. The gate was not changed. The same worker's compiler warnings showed two truncations in the certification report (`System_HeH_mol_metals` printed as `System_HeH_mol_metal`, the carrier row's scale text cut at 76 characters); both fields were widened (log text only). The diagnostic shares the gate's reduction with the verdict (`species_row_distance_over_gated_cells`); the certification suite passes 143 of 143, and with the variable unset every output is byte-identical to the build before it (MEASURED by the worker on four states).

Item 6, a grid option that keeps the interior. Outer shells `[r_face,cells]: <r_face> <cells>` (`K33d` of `md/input_schema.md`, default off): the grid is built as before, and `<cells>` cells are appended beyond its outer face, their widths in geometric progression with the ratio that puts the last face exactly on `r_face`; every constructed center up to the first outer ghost, every face, width and volume and the window indices are bitwise those without the key (new test `grid_and_gates/outer_shells_base_identity`, 40 of 40), N and everything sized on it follow the extended grid, and a state written with shells carries them in its `# grid` line, so it loads only into a run with the same shells. With the key absent every output is byte-identical to the control build (MEASURED on an LHS 1140 b evaluate and on `hydrostatic_column`).

Item 5, the domain test (P1.4 (c)) and the outer closure (`md/lhs1140b_outer_shells_domain_test_20260924.md`).

The straddling molecular states on the 500-cell grid plus 43 shells to 60 `R_p` (spacing ratio continuous to $4e-4$) both certify (13 and 10 passes). MEASURED: the wind turns transonic, with a sonic point at 40.3 `R_p` (Mach 0.51 at the old 29 `R_p` edge becomes 0.77); the EW falls by 1.74 per cent at both compositions (1.0964 to 1.0773, 1.1532 to 1.1330), $\log_{10} \text{Mdot}$ moves by $2e-4$, and the profiles move by at most 0.13 per cent inside 2 `R_p` and by up to 20 to 30 per cent at the old edge. The EW crossing moves from He/H 1.4643 to 1.4886; with the matched kernel (unchanged, 22.2 km/s) both extended states lie below 1.108. The H2 carrier's last-cell defect stays two decades below the gate

on both domains ($5.6\text{e-}7$ at $60 R_p$): the zero diffusive outer face does not decide these states; the subsonic outflow boundary at $30 R_p$ holds the outer wind on a subsonic branch and acts through the hydrodynamics (tentative).

Item 7, the composition update holding the pressure. Composition update holds: energy|pressure (K15j, default energy): with pressure, the composition update of the alternation keeps ρ , v and p of every cell, re-closes the eliminated species at the temperature of the new composition and rebuilds the energy from the caloric equation of state (rebuild_energy_at_held_pressure, EXHALE_main.f90), printing the energy it adds. MEASURED (md/lhs1140b_composition_update_pressure_20260924.md): it removes what it was built for, the base pressure step (cells 1 to 3 held exactly; the base face mass flux handed to the next solve $4.46\text{e-}8$ against the wind's $4.44\text{e-}8$, where the default hands $-2.84\text{e-}5$), and the next hydrodynamic solve still fails in all three tests: on the well-mixed He/H 0.083 and 0.55 cases on the energy row at the H2 front (the carrier update raises the cooling there by 31 to 44 per cent at the fixed wind, under either value), and on the $K_{zz} = 1\text{e}9$ rung 1.8 to 1.6, where the default recovers and certifies at pass 21, it fails from pass 2 on and never recovers. The obstacle behind the base step is the front's energy balance under a composition change at fixed wind. The default is unchanged; the option stays, default off, with its negative result recorded. With the key absent every output is byte-identical to the control build (MEASURED on the P1.6 entry and on hydrostatic_column).

Item 8, one route for every molecular catalog case. make_models.py wrote Restart intent: stationary equilibrate for a molecular case; the certified molecular cases carry stationary, and were solved under it from an atomic-to-molecular conversion as well as from molecular continuations (He/H 2.13 and 9.7, runs of 2026-09-19, READ). So instead of a rule by the kind of seed (the recommendation), every molecular case now takes stationary: make_models.py writes it, with Coupled carrier solve: False and the base-grid comment line the certified cases carry, and its output now equals the input.inp of every molecular $K_{zz} = 1\text{e}9$ and well-mixed case byte for byte (the two photochemical molecular cases differ only in the order of one line); molecular_scalar_gj1132_kzz1e9/HeH0.083/input.inp, the one case with equilibrate, was changed to match.

Item 9, the On stall handover: deferred.

Item 10, the goldens. Refreshed for the sixteen cases the matrix of section 12 ran (all but oxygen_chemistry), from the outputs of that matrix; the previous goldens are kept in backup/regression/golden_pre_cfl_20260924/.

Item 11, oxygen_chemistry. The case now starts from its own golden of 2026-09-20 (the 1000-step snapshot of the cold start, kept in its IC/) and marches 1000 steps from there (Load IC? True, Restart intent: relaxation; its README.md records why); the load passes the ghost fixed point and the run ends at $du = 20.9$, $\log_{10} M\dot{=} 9.39$ (MEASURED with the control build). Its golden was snapshotted from that run.

The smaller items. The Phase-E overlap window is entry (AH) of md/TO_BE_DONE.md, deferred to the next time the LHS 1140 b profile runs are handled. docs/Update_EXHALE_stage2.{tex,pdf} was regenerated for the edit of its section 17 (310 pages, no error). src/utils/map_state_to_grid.py wrote nan as the center shift in its # mapped: line between grids of different cell counts; it now says the shift is not defined there.

15. The grid of the crossing (decision item 3) and the regression of sections 13 and 14 (2026-09-24)

Three grids, three compositions. The certified molecular $K_{zz} = 1e9$ states at He/H 1.45, 1.52 and 1.6 were mapped onto grids of 1000 and 2000 cells (the base block and every other spacing halved at each step, the outer radius unchanged; target grids from cold-start dumps) and solved again with the rung environment; all six certify (1000 cells: 16, 15, 15 passes; 2000 cells: 13, 13, 19). MEASURED through the catalog's evaluation, synthesis and EW operator (LHS1140b/models/.P1/refine_grid/, .P1/connectivity/grid_convergence.txt):

He/H	EW, 500 cells	1000 cells	2000 cells	continuum (DERIVED)	log10 Mdot 500 / 1000 / 2000
1.45	1.0964	1.0640	1.0475	1.030	7.85 / 7.84 / 7.84
1.52	1.1532	1.1196	1.1025	1.085	7.86 / 7.85 / 7.85
1.6	1.2155	1.1804	1.1625	1.144	7.86 / 7.86 / 7.86

The EW converges at first order (successive-difference ratio 1.96, $p = 0.97$, at every composition), and the continuum column is the Richardson extrapolation at that order. The crossing of the working target 1.108 moves from He/H 1.4643 (500 cells) to 1.5054 (1000), 1.5273 (2000) and 1.552 (continuum); from the broadened line of section 13, from 1.5112 to 1.5547, 1.5780 and about 1.60 (the last just above the 1.6 rung); every other value is interpolated between certified rungs. The matched kernel does not move with the resolution (22.18 to 22.21 km/s on all nine states). The 500-cell catalog overestimates the EW by about 6 per cent, twice the observation's working error; a formally second-order scheme converging at first order in the EW points to a first-order error in the line-forming region, whose source is not established. Combined with the domain factor of section 14 (DERIVED, the two measured separately), the crossing stands near 1.58 from the EW and near 1.63 from the broadened line. Which grid the catalog should use is left to the user with these numbers.

The regression of sections 13 and 14. After all the code of this series had landed (the diagnostic of item 4, the grid option of item 6, the composition option of item 7, the widened report fields, and the mapper record), the tree was rebuilt (EXHALE.x md5 2df491cb020e8200be247e2469964a63, no warning) and the whole matrix run against the goldens refreshed in section 14: REGRESSION PASS, all seventeen cases byte-identical (66 files; the two _adv files of iontrans_metals have no reference, as that route writes none), oxygen_chemistry included on its warm start.

16. The user's decisions of 2026-09-24 on the four remaining items

Put to the user after sections 14 and 15; decided the same day.

The catalog's grid: 500 cells, the compositions quoted from the grid continuum. The catalog keeps its 500-cell grid (every certified state, the chains and the ladders stay as they are); a composition read from it is quoted as the continuum value of section 15, with the difference between the 500-cell and the continuum value as the grid uncertainty of every catalog number: from the EW, He/H 1.55 (the 500-cell crossing is 1.46, 0.025 dex lower, because the 500-cell EW is about 6 per cent high), from the broadened line about 1.60 (1.51 on 500 cells); with the domain factor of section

14, 1.58 and 1.63 (DERIVED, the two effects measured separately). docs/lhs1140b_catalog_state.tex section 5 states the rule.

The gating candidate: not adopted. The species rows keep their current gate; the diagnostic EXHALE_CERT_GATE_DIAGNOSTIC=1 stays as a default-off report.

The pressure-holding composition update: kept, default off. Composition update holds: pressure stays in the code with its negative result (section 14, item 7); the default remains energy.

The energy imbalance at the H2 front: explored. The obstacle item 7 exposed, the energy row at the H2 front failing after a composition change at fixed wind, is being diagnosed (LHS1140b/models/.P1/h2_front_energy/); its result is recorded in a later section.

17. The atomic-data audit of 2026-09-24, applied

md/EXHALE_ATOMIC_DATA_AUDIT_2026-09-24.md (an audit of the atomic data against the sibling code MoCHII) was checked item by item in the code and, where it could be obtained, in the original source, and applied; the four memos carry every verdict, source, size and command: md/atomic_data_audit_applied_core_20260924.md (the H/He cooling core), md/atomic_data_audit_metal_tables_20260924.md (the CHIANTI metal tables), md/charge_exchange_detailed_balance_20260924.md (the charge-exchange table) and md/atomic_data_audit_applied_docs_20260924.md (comments, unreachable code, the detailed-balance test). Every item the audit raised was found to be real; the corrections are physics corrections and are on by default.

The H/He cooling core.

- Recombination cooling of every ion is $k T \alpha (3/2 + d \ln \alpha / d \ln T)$ on the coefficient the balance itself uses (B1, B7, B12). Hui & Gnedin (1997) print the He III fit with a factor 8 where the hydrogenic scaling gives 2: He III removed 3.05 kT per capture at 1e4 K and now 0.76 kT. The relation reproduces Hummer (1994) Table 1 to 0.1 per cent.
- The post-process temperature root and eval_cool now call one cell assembly, radiative_cooling_of_cell (B2); the two coolings differed by a factor 1.04-1.60 on the wasp_full post-process and now agree to 1.3e-14.
- **The $2^3S \rightarrow 1^1S$ electron-impact de-excitation (B3)**, the detailed-balance reverse of q13, is in every place the triplet balance is written, with the 19.82 eV returned to the gas. Oklopčić & Hirata (2018) leave it out as small against $2^3S \rightarrow 2^1S$, 2^1P ; that holds at 1e4 K (ratio 0.055) and not below: 0.15 at 5e3 K, 0.57 at 3e3 K, 2.8 at 2e3 K (MEASURED), the temperatures of the LHS 1140 b line-forming region.
- Voronov (1997) fits take Voronov's fitted dE again (B4: Mg II x1/1.47, C I x1/1.10 at 5e3 K); n = 2 recombination is the balance's case-B coefficient split with Draine's 2s fraction (B6, +2.6 per cent at 1e4 K); He I, He II (A3) and H I (B8) collisional-excitation cooling are CHIANTI 11.0.2 sums with the observed level energies (x1.11, x1.29 and x1.11 at 1e4, 1e4 and 5e3 K), H I on one rate set shared with the n = 2 model; single-precision literals are gone from the four files (B10); use_2lev_cool, lambda_2level, melem_Z, cool_FeII_func and the other unreachable wrappers are removed (B11).

The metal line cooling (A2). Mg II took the CHIANTI .scups scaling energy (4.27 eV) as its transition energy where the observed g-weighted 4.4300 eV belongs, 1.15 times too high at 1e4 K and 1.39 at 5e3 K; it is refitted with observed

energies and every bound channel (0.2 per cent). Ca II and Mg I carried the resonance line alone; they now take (T, n_e) statistical-equilibrium tables with every channel (at $n_e = 1e9 \text{ cm}^{-3}$ the old coefficient was 1.7-4 times low for Ca II and 5.8-480 times low for Mg I over $1e4\text{-}3e3 \text{ K}$). Fe I stays on its Boltzmann (LTE-limit) table, now stated as an upper limit at most 2.0 times the low-density value; the data for a density-resolved Fe I solve are not available here.

Charge exchange. The reverse rows of Huang et al. (2023) Table 4 depart from detailed balance with internal partition functions by factors 0.18 to 2.9. Each measured or computed direction is kept as the original paper gives it, and the other is computed from it by detailed balance with NIST partition functions; four reverse rows are not carried (A2, A10, C4, C6), their forward channel ending in an excited or radiative product; three transcriptions are corrected to the originals (A12 $\text{Si}^{2+} + \text{H}$: $1.23e-9$ and the sign of its exponent; C5: Zhao et al. 2004 eq. 19 in full; A15: Stancil et al. 1998 Table 1 directly). Ten pairs whose originals could not be obtained (mostly Physical Review A) stay as printed and are marked KNOWN_BLOCKED in the new test `src/tests/charge_exchange_detailed_balance`; among them $\text{N} + \text{H}^+$ (its endothermic direction falls with T) and $\text{Na} + \text{H}^+$ (both directions below $1e-60 \text{ cm}^3 \text{ s}^{-1}$, so no Na-H charge exchange acts).

Other corrections of the same series. Comments that contradicted the code or the source (C1, C4 to C9, C11) are corrected; `rk_R21_H_Hep_cx`, `rk_R22_Hp_He_cx` and the three unreachable Lyman-Werner fits are removed after a call-graph search; the Lyman-Werner warning of `ionization_equilibrium.f90` and `write_output.f90` no longer cites a DB96 fit the band share has not used since 2026-09-06 (`NH2_db96_max` removed); `src/tests/a2_m1` compiles again against a built tree and its saved output was regenerated (the 0 K photolysis thresholds; `md/a2_reaction_audit.md` carries a correction block); the Makefile flag stamp now hashes the first line of `$(FC) --version`, whose conda-forge package build number `-dumpfullversion` drops: the previous build/ held 35 objects of gcc 16.2.0-4 and 92 of 16.2.0-5 linked into one binary.

Checks. The tree was rebuilt from `make distclean` (EXHALE.x md5 551dae520985d63c9ea8b15c21106589, no warning; copied to `LHS1140b/models/EXHALE_551dae52.x` with `BINARY_MANIFEST_551dae520985.txt`); `physics_probe`, `charge_exchange_detailed_balance` (65 pass, 10 known blocked) and `charge_exchange_rows` pass on it. The regression matrix and the remaining suites are recorded in the next sections.

Effect (MEASURED by the workers, on builds of the same physics). `wasp_full`: $\log_{10} \text{Mdot}$ $-2e-4$ (cooling core), $+4e-4$ (charge exchange), $-2.7e-3$ (metal tables, a du-stopped case, so below its path spread); Fe III $x_{0.48}$ at 1.2 R_p . LHS 1140 b: the He 10830 EW of the certified molecular states falls by 14.9 to 16.7 per cent on evaluation, almost all from B3, and by 20 per cent after one stationary re-solve (He/H 1.45: 1.0964 $\rightarrow$ 0.8775 %A; 1.52: 1.1532 $\rightarrow$ 0.9180), because $n(2^3\text{S})/n(\text{He})$ falls by 41-58 per cent at 5-10 R_p , where n_e q_{31g} is 3-100 times A31. No certificate of the catalog holds under the corrected physics. **Every LHS 1140 b number of this log before this section, and of `docs/lhs1140b_catalog_state.tex`, is of the physics before the audit.**

18. The H2-front energy imbalance, the H3+ non-LTE interpolant, and the descent below He/H 1.45 (2026-09-24)

The H2 front (decision of section 16). Diagnosed with the tree binary of section 15 and no change to the source (`md/lhs1140b_h2_front_energy_imbalance_20260924.md`, figure

docs/figures/lhs1140b_h2_front_energy_imbalance.pdf). MEASURED: across the composition update one term of the front's energy row moves, the H3+ infrared cooling (99.6 per cent of the cooling there): on the well-mixed He/H 0.083 update H2 at the front rises 3.2 per cent, H3+ answers with +8.6 per cent (+17.5 with the pressure held), and the cooling of the spike cell rises by 31 (44) per cent against a heating that rises by 1 per cent. The code path is consistent (the next solve assembles the relaxation's cooling to 1e-7 and 3e-5). About two thirds of the jump came from the interpolant of the H3+ non-LTE factor (below). With the composition frozen a hydrodynamic root exists and the trust-region solve reaches it ($\|R\|$ 1.5e-7 and 1.6e-8), where the line search cannot take its first step; the well-mixed He/H 0.083 alternation still does not converge with the pressure held and the trust region (6 passes, the front walking outward, $\dot{M}$ -26 per cent). The trust region inside the alternation is a solver-control change and was not made.

The H3+ non-LTE factor is now a power law between the columns of Miller et al. (2013) Table 6 (h3p_cooling.f90, s_table). The emission per H3+ ion can grow at most as fast as the collision rate that populates the emitting levels, so the local index $d \ln s / d \ln n(\text{H2})$ lies between 0 and 1, and the table's node-to-node indices do (0.00 to 0.89). The interpolant linear in s over $\log n$ gave 12.6 just above the $1\text{e}8\text{ cm}^{-3}$ column and 0.17 just below it at 2000 K (DERIVED), and over $r = 1.25\text{--}1.40 R_p$ of the well-mixed He/H 0.083 state an H3+ cooling 1.8 to 5.6 times the power law's. In T the interpolant stays linear; its form there is not fixed by any bound and is written at the code with the row ratios. The correction is on by default and is in the binary of section 17.

The descent below He/H 1.45 (decision item 2 of section 14) ran under the P1.2 rules on EXHALE_46d5b9f4.x (the physics before section 17; LHS1140b/models/.P1/descend_from_1.45_record.txt): 1.152 from 1.45 not certified (18 passes, 3 h); 1.2924 from 1.45 certified (16 passes); 1.152 from 1.2924 not certified (24 passes, the H2 carrier row $3.1\text{e-}4$ at cell 247); 1.2202 from 1.2924 not certified (24 passes, the He/H element row $8.6\text{e-}5$ at cell 234 and falling by about 10 per cent a pass); the driver stopped on two consecutive failed refinements. The interval 0.55 to 1.29 stays unsampled, and the one new certified state is of the physics before section 17.

The molecular family under the corrected physics (user decision of 2026-09-24: this family first). The certified He/H 1.45, 1.52, 1.6, 1.8 and 2.13 states are being re-solved from themselves with EXHALE_551dae52.x (LHS1140b/models/.P1/newphys/, the rung environment, 24 passes, 4 h); their result, the new crossing and what follows for the rest of the catalog are recorded in a later section.

19. The second round of the atomic-data audit (2026-09-24/25)

The audit's author rechecked the applied tree from MoCHII (~MoCHII/MoCHII_v1.00/md/EXHALE_AUDIT_RECHECK_2026-09-24.md and ..._SECOND_RECHECK_2026-09-24.md): every item of section 17 was confirmed as applied correctly, and the open ones were listed. Each was verified in the code and the original source before it was applied; memos md/atomic_data_audit_round2_core_20260924.md, md/charge_exchange_detailed_balance_20260924.md section 10 and md/chianti_descaling_spline_20260924.md.

Recombination and the helium network.

- The ground-state capture coefficient α_1 is the Milne relation on the photoionization cross section the transfer uses, for H I, He I and He II (0.998 of Hummer 1994 for H I, 1.006-1.011 of Hummer & Storey 1998 for He

- I); case B = Badnell alpha_A minus that alpha_1 (the Mao & Kaastra fit, 2.7 per cent low for He I at 1e4 K, is removed).
- The Oklopčić & Hirata (2018) alpha_1 is the capture to 1¹S alone; with the singlet/triplet coupling off the excited-singlet captures were missing. Case B is now split into triplet and singlet captures with Hummer & Storey (1998) Table 5 in both configurations.
 - **H ground captures escape where the gas is thin** (new key H_rec_escape, default True, K14d2 of md/input_schema.md): alpha_HII = alpha_B + y_HI alpha_1 with the local escape weight of the He II construction, case A in the thin outer wind and case B in the thick base; False is case B everywhere. A correction of the case-B assumption where it does not hold, hence on by default.
 - The He III recombination cascade (ground capture, direct capture to n = 2, He II Ly-alpha, the two-photon continuum of Nussbaumer & Schmutz 1984, the 2s-2p mixing of He II) is absorbed on the spot under He_rec_coupling.
 - The collisional feed of 2³S through the higher triplets (feed/q13 = 0.018, 0.101, 0.328 at 5e3, 1e4, 2e4 K, CHIANTI 11.0.2) is added to q13 wherever the triplet balance reads it, and so are the 2³S -> n¹L (n >= 3) excitations.
 - A(2s -> 1s) = 8.2249 s⁻¹ (Nussbaumer & Schmutz 1984; was 8.26); the 2s-2p l-mixing of H by protons, He+ and He2+ from Pengelly & Seaton (1964) with their own cut-off rule (the Lamb shift or fine structure for 2s, not the Debye radius); the 2³S electron-impact ionization as Black (1981) eq. 12 prints it; coll_rate_prefactor = 8.629132e-6 from CODATA, one parameter; the 2³S triplet cooling is a CHIANTI sum (Black's 13179 K was his fitted threshold, not an energy); the (T, n_e) cooling tables above their top density continue with the last slope clipped to [-1, 0]; q31g is recorded in the diagnostic states (state 28 of stage_channels.txt).

Energy terms that were missing. The metal recombination energy ($k T \alpha (3/2 + \ln \alpha / \ln T)$) and the metal collisional-ionization energy (potential times the Voronov rate) are in the cooling; the charge-exchange energy defects heat the gas (new heat channel 20, heat_charge_exchange: per reaction the defect less the energy radiated by an excited product, product states from the sources read; the ledger closes to 1.4e-8 eV an event). On wasp_full the charge-exchange heat is 18 per cent of the heating at the base, 3.9 at 1.1 R_p, 0.4 at 1.3 R_p; the base share is mostly Fe + H+, whose product state could not be read (Rutherford & Vroom 1972) and is taken as ground, an upper bound.

Charge exchange. N + H+ / N+ + H are now Kingdon & Ferland (1996) Table 1 (Butler & Dalgarno 1979) with the reverse by detailed balance (the printed Huang rows were 69 and 1.5e7 times low at 1e4 K); Na + H+, Na+ + H and K + H+ are not carried (no reading of the printed Table 4 form gives a physical rate; the originals, Dutta et al. 2001 and Watanabe et al. 2002, could not be obtained); D26 and D30 are derived by detailed balance from the source directions; the gate now also requires every carried rate to obey the threshold law (205 pass, 6 known blocked, 2 known defects).

CHIANTI descaling. CHIANTI prescribes the natural cubic spline through the .scups points (descale_scups.pro, CHIANTI Technical Reports 4 and 13); the scripts of cooling_data/ used the not-a-knot spline (as ChiantiPy). Every CHIANTI-based coefficient was regenerated with the natural spline: He I cooling moves by -2 to +0.5 per cent over 3e3-3e4 K (Upsilon(1¹S-2¹P) by -16 per cent at 1e4 K), H I by +/-0.7 per cent, O I fine-structure Upsilon by +1.3 to +6.7 per cent, most others below 1 per cent.

Other repairs of the series. The Lyman-Werner band share the H2O/OH continua see is now summed cell by cell

along the path at each cell's own (T, n_H) (it was one uniform slab at the local T: the band's photon ledger was 18 per cent short on the warm-started oxygen_chemistry state; only Oxygen chemistry: True runs reach it); six test suites were repaired (md/test_suite_repairs_20260924.md).

Build and checks. Tree rebuilt from make distclean: EXHALE.x md5 b059a4be0459d65fdc3dff9894ba5d2f, no warning, copied to LHS1140b/models/EXHALE_b059a4be.x with BINARY_MANIFEST_b059a4be0459.txt. All 42 suites run on it (MEASURED): 38 pass; grid_and_gates fails its three stationary_evaluate_* rows on the pinned fixture backup/regression/wasp_full_newton/IC/ (solved under the old physics; a re-solve under this physics certifies, info = 0, ||R|| 2.3e-9, and waits in LHS1140b/models/.P1/suite_repairs/wfn_IC_r2/ for the golden refresh); residual_determinism fails its known closure-spread row (1.17e4, the solver item open since 2026-09-18); the two steady_* suites need the log of a finished steady solve as input.

Effect (MEASURED by the workers). Of the core items alone: wasp_full log10 Mdot 13.35 -> 13.32, the H ionization front outward (x(H⁺) 0.36 lower at 1.27 R_p), T up to +11 per cent; mol_metals -0.004. Of the charge-exchange items: wasp_full +4.2e-3, N II down by up to 1.8 dex. On the LHS 1140 b He/H 1.45 state the core items lower the He 10830 EW by 7 per cent on evaluation, mostly through the triplet/singlet split of case B (0.041 %A), H_rec_escape (0.020) and the Milne alpha_1 (0.011).

The LHS 1140 b molecular family between the two rounds (binary EXHALE_551dae52.x, LHS1140b/models/.P1/newphys/): re-solved from the certified states, no rung certified in 24 passes (the H2 carrier row at the front falling by 0.72-0.86 a pass), then all five certified in 7-11 more passes. Their EWs (MEASURED, evaluate + synthesis + the EW operator): He/H 1.45 0.9813, 1.52 1.0185, 1.6 1.0587, 1.8 1.1500, 2.13 1.2765 %A; the 1.108 crossing near He/H 1.70. These are of the physics of section 17 only; the states under this section's physics are being solved from them.

20. The molecular K_{zz} = 1e9 family under the audited physics (2026-09-25)

The user's decision of 2026-09-24 was to re-solve this family first. The five states He/H 1.45, 1.52, 1.6, 1.8 and 2.13 were re-solved from the states certified under the first round (section 19) with EXHALE_b059a4be.x (LHS1140b/models/.P1/newphys/HeH<x>_r2/, the rung environment, resolve.sh): all five certified in 13 passes (the H2 carrier row at the front falling from 0.5 to 6e-6 - 9e-6). Each was then evaluated, synthesized and measured with the EW operator (evaluate_ew.sh, eval_r2/; every evaluation certified). MEASURED:

He/H	EW, audit round 2	round 1	before the audit	log10 Mdot, round 2
1.45	0.9387	0.9813	1.0964	7.899
1.52	0.9776	1.0185	1.1532	7.906
1.6	1.0198	1.0587	1.2155	7.913
1.8	1.1159	1.1500	1.3584	7.927
2.13	1.2497	1.2765	1.5604	7.945

The 1.108 crossing of the log-log chord moves from He/H 1.4643 (before the audit) to **1.7833, between the certified 1.6 and 1.8**, on the 500-cell grid; the EW falls by 14 to 20 per cent at fixed He/H and log10 Mdot

risers by about 0.05. The grid continuum factor and the domain factor of sections 15 and 14 were measured under the old physics and are not re-measured here. Figure docs/figures/lhs1140b_kzz1e9_audit_stages.pdf (.P1/newphys/compare_physics_stages.py): the EW against He/H at the three stages, the line at He/H 1.8, and T, v, n(2³S), x(H⁺) and x(H₂) at He/H 1.8. At He/H 1.8 (MEASURED on the _adv profiles, before the audit / round 1 / round 2): the line keeps its shape and loses depth; the peak temperature is 5439 / 5315 / 5600 K; the velocity at the outer edge 1.21 / 1.40 / 1.36 km/s; x(H⁺) at 20 R_p 0.059 / 0.056 / 0.050 (the ground captures escape in the thin far wind); the metastable density at its peak (1.35-1.38 R_p) 33.3 / 33.3 / 31.8 cm⁻³, and its radial integral over 1-1.5, 1.5-3 and 3-30 R_p 11.1 / 10.9 / 10.2, 28.8 / 28.3 / 27.6 and 28.4 / 24.2 / 23.4 cm⁻³ R_p. The loss is in the cool outer wind beyond 3 R_p (-18 per cent), where the superelastic 2³S → 1¹S collisions outrun the radiative decay, and the EW falls by 18 per cent with it.

21. The code-size appendix moves to this log (2026-09-25)

On the user's instruction the hand-written code-size appendix docs/Update_EXHALE_appendix.tex is now input by this log's typeset twin instead of the stage 2 twin (src/utls/update_log_to_tex.py, STAGES); docs/Update_EXHALE_stage2.{tex,pdf} were regenerated without it. Its references to "this log" now name the stage 2 log where they meant it, and it states that its numbers (measured 2026-09-22) do not yet include the code of stage 3; they are re-measured with src/utls/codesize.py when the atomic-data series is closed.

22. The originals supplied on 2026-09-25, read and applied

The user supplied the papers the two audit rounds could not obtain; each was read and the code checked against it (memos md/charge_exchange_detailed_balance_20260924.md sections 11 and 11.8, md/atomic_sources_hhe_20260925.md).

Charge exchange.

- Na + H⁺ (Dutta et al. 2001, non-radiative, product H(n=2), endothermic by 1.682 eV) and Na, K + H⁺ (Watanabe et al. 2002, radiative): neither paper gives a fit and Huang et al.'s Table 4 forms are in neither; the rate is now the Maxwellian average of Dutta's cross sections (their own rate figure violates the threshold law below about 5000 K) plus Watanabe's radiative values: 8.4e-12 at 1e4 K against 4.4e-171 as coded before. The reverse is not carried (it starts from H(n=2)). K + H⁺ from Watanabe.
- Fe + H⁺, O⁺, N⁺ (Rutherford & Vroom 1972 Table I): the 300-1200 K values in log-log, held above 1200 K; the measured cross sections start at 8 eV, so every wind rate is an extrapolation, bracketed by 1.4e-9 and 4.0e-9 at 1e4 K (DERIVED; the held value is the source's own extrapolation to 6 per cent; Pequignot & Aldrovandi 1986 infer 2.7e-9). The product state is not determined in the paper; the near-resonance reading (Fe+ z4F/z4D, 1.18 eV of heat an event instead of the full 5.70 eV defect) is the one Pequignot & Aldrovandi's Table 3 also takes; the range is stated at the code.
- S + H⁺ (Zhao et al. 2005 Table III), Si + He⁺ (Satta et al. 2013 eq. 8, Table 1), S + C⁺ (Chenel et al. 2010 Table 2), Na + Ca⁺ (Kwolek et al. 2019, measured at 0.015-0.09 eV, the Langevin value kept as an upper bound) replace the Table 4 transcriptions; the reverses of channels whose forward ends in an excited product (Fe rows, Si

+ He+, Na + Ca+) are not carried. C + Si+ (D1/D2) stays blocked: the paper Table 4 cites for it does not contain it. The detailed-balance gate: 194 pass, 0 fail, 1 known blocked.

- Effect: `wasp_full` $\log_{10} \dot{M}$ -3e-4; the charge-exchange heat at the base falls from 18 to 6.7 per cent of the heating; Na I D2 transit depth 1.169 -> 1.156 per cent.

Hydrogen and helium.

- The He I collision strengths of the metastable network (q13, q31g, q31a, q31b) were exponential fits the ATES code carried, not coefficients of Lampon et al. (2020), who take the values from Bray et al. (2000). Bray tabulates 5.6e3-5.6e5 K; the fits passed through the first four table values and were an unsupported extrapolation below 5.6e3 K (1.10-1.16 times the new strengths at 1.5e3 K). Now: Bray's table inside its range (PCHIP in log-log; one misprint of Table 2 corrected), the CHIANTI `he_1` temperature dependence joined continuously outside it; the feed through the higher triplets, the 2³S cooling, the 2³S -> n¹L excitations and the He I cooling sums were rebuilt on Bray. On the LHS 1140 b He/H 1.8 state the He 10830 EW rises by 2.3 per cent on evaluation.
- H 2s-2p mixing by electrons: the coded rate (Janev et al. 2003, from Chibisov 1969) was ten times below Seaton's (1955) first-order dipole constant; now Seaton (1955) eq. (55), approximation V (5.78e-5 cm³ s⁻¹ at 1e4 K against his Table 1 5.7e-5; Osterbrock & Ferland Table 4.10 agree).
- A(2s -> 1s) = 8.22461 s⁻¹ (H) and 526.823 s⁻¹ (He II), Drake (1986) eq. (27) (section 19 wrote 8.2249, the Nussbaumer & Schmutz 1984 value it replaced).
- The case-B share of recombinations into H 2s is now Pengelly's (1964, Table I) hydrogenic fraction, the one the He II branch already used; the quadratic in T that section 17 (item B6) called "Draine's 2s fraction" is Christie et al.'s (2013) fit, and Draine (2011) gives only Tables 14.2 and 14.3.
- The atomic-mode (no 2³S) He I case-B photons are now sent at their own exits (584 Å, 19.8 eV, two-photon) split by the network's rates, instead of one averaged energy (3.10 instead of 4.12 eV of heat a capture at high density; z = 0.966 and 0.660 at low and high density against Draine's 0.96 and 0.67).
- Voronov (1997): all 20 rows equal the published Table I; its stated range (1 eV-20 keV) is written at the code with what the fit does below it. Anderson et al. (2000) confirm the CHIANTI H I strengths to 2.5 per cent at 0.5-5 eV; Yan & Babb (2023) published values equal the arXiv ones; Dere et al. (1997) and Del Zanna et al. (2015) say nothing about the descaling spline, which rests on the CHIANTI IDL code.

Build. Tree rebuilt from `make distclean: EXHALE.x md5 50aa18d79bcd25f219b5b59baa31b8b8`, no warning, copied to `LHS1140b/models/EXHALE_50aa18d7.x` with its manifest. The manifests of 2026-09-24/25 listed files as changed that were not (the header lines of the previous manifest matched the file-name search); the three lists were rebuilt from the md5 lines of consecutive manifests. The refined-grid runs of the round-2 physics that were still running (He/H 1.6 on 2000 cells, 2.13 on 1000 and 2000) were stopped as superseded; the molecular family, the regression fixture, the regression matrix and the suites are being re-run on this binary.

The texts the corrections made stale (`md/stale_texts_20260925.md`): the physics document (`docs/physics_overview/*.tex`, the PDF rebuilt, 70 pages), `docs/transmission_spectrum.{tex,md,pdf}` and `md/atomic_data_EXHALE_vs_MoCHII.md` were brought to the code. The transit post-processor built the n = 2 populations of the Balmer lines with Christie et al.'s rates while claiming to mirror the solver; it now uses the solver's

rates, one transcribed function each (agreement with the Fortran 3.1e-13, MEASURED), and keeps only its own Ly-alpha and Balmer fields (which the profiles do not carry). H-alpha peak depth of HD 209458 b -1.73 per cent, of wasp_full +0.14 per cent (MEASURED), almost all from the proton l-mixing, which lowers $n(2s)/n(2)$ at 1.3-3 R_p from 0.53-0.69 to 0.26-0.34. The $n = 2$ threshold cross section $1.4e-17 \text{ cm}^2$ is Seaton (1959) at the $n = 2$ edge ($1.378e-17$ DERIVED), not Osterbrock & Ferland.

23. The Markdown documents move to md/ (2026-09-25)

On the user's instruction the 350 Markdown files directly under docs/ moved to the new top-level md/; docs/ keeps the LaTeX and PDF documents, the figures and the dated audit directories, whose 13 Markdown files stayed with the files they describe. Done with mv, not git mv: git sees the 35 tracked files as deleted under docs/ and new under md/. Paths that followed:

- .gitignore: the rule that keeps the development record off the public repository (docs/*.md with 35 exceptions) is now md/*.md with the same 35 exceptions under md/.
- 73 relative links of 24 moved files into docs/ subdirectories now read ../docs/...
- 3079 references docs/<name>.md in 416 maintained files (sources and their comments, tests, scripts, README.md, README_HOWTO.md, the md/ files themselves, the docs/ LaTeX, the regression README and scripts) now read md/<name>.md; four of them are strings the binary prints (the setup report, one lower-atmosphere message, one test message), so the tree binary changed (EXHALE.x md5 b30e56f63335431dcb22b9fe3f80f948, no warning); the state files the regression compares carry none of them.
- src/utis/update_log_to_tex.py reads the logs from md/.
- Left as they were: run records and products (inputs, outputs, state files, REPRODUCE.md, logs, the .P1 run areas, archived regression snapshots), whose text records the paths of their day and whose bytes other records hash; and the one string LHS1140b/models/make_models.py writes into the 94 stored catalog inputs, so that it still reproduces them byte for byte.
- The workspace CLAUDE.md and the session memory point to md/.
- The 17 docs/ documents whose text names these paths were rebuilt on the user's instruction (the two log twins regenerated first): every build without error or undefined reference, the new paths in the PDFs.

24. The last items of the MoCHII comparison, and the molecular crossing under the corrected physics (2026-09-25/26)

md/EXHALE_MOCHII_ATOMIC_COMPARISON_2026-09-25.md (a side-by-side of the atomic processes of EXHALE and MoCHII, written from the MoCHII side) found the recombination construction, the recombination cooling, the He 2^3S network and the H 2s-2p mixing now the same in both codes to four digits at $1e4 \text{ K}$, and listed five open items for EXHALE; each was verified and applied (memos md/charge_exchange_detailed_balance_20260924.md section 12 and md/photoionization_584_auger_20260925.md).

- $N2^+ + H \rightarrow N^+ + H^+$ (Barragan et al. 2006 Table 2, $9.70e-10 \text{ cm}^3 \text{ s}^{-1}$ at $1e4 \text{ K}$, the product $N^+ 2s2p^3 3D0$,

so no reverse; heat 4.57 eV an event; metals.inp: cx_N2p_H <scale>, default 1). On wasp_full N III falls by up to 5.4 decades.

- **He2+ + H -> He+ + H+ + photon** (Kingdon & Ferland 1996 Table 1, 1.0e-14, radiative, no reverse and no heat), in every system that carries He III; He III falls by 0.69 dex at the wasp_full base. The original of the rate (West, Lane & Cohen 1982, Phys. Rev. A 26, 3164) could not be obtained.
- **Density-dependent detailed balance.** Where the source names the product level, the reverse is the thermal reverse times the population-weighted factor of the metastable levels at the local (T, n_e) (five-level statistical equilibrium of N I, S II, C I from CHIANTI 11.0.2), and the heat of the forward channel removes the forbidden-line energy a metastable product radiates at low density; elsewhere the thermal reverse is exact at every density. wasp_full (n_e 8e8-3.5e9) moves by less than 0.2 per cent. A 0/0 below 60 K in the metastable Boltzmann weights (one LHS 1140 b base cell sits at 40.8 K) was guarded.
- **He I 584 A.** The line is resonantly scattered by He0 inside the cell, and each scattering ends in absorption by H I (or H2, or a metal ion), in conversion to 2^1S through the 2.06 um branch (1/912) and the existing two-photon exit, or in escape (Wood, Mathis & Ercolano 2004 sect. 4.1, eqs. 15-17, generalized to every continuum absorber of the cell and to escape); their eq. 17 carries He/H = 0.1 in its coefficient.
- **Inner-shell photoionization.** Each subshell absorption heats with the photoelectron plus the Auger energy, the fluorescence and the further ionization energy are not heat, and the neutrals of C, N, O, Mg, Si, Ca and Fe are carried straight to X++ where the vacancy autoionizes (Kaastra & Mewe 1993 Tables 2 and 3; Ca II 3s is treated as radiative, since the measured energies forbid its autoionization). At 98 eV the electrons of a Mg II absorption receive 0.333 of the photon energy instead of 0.847.

Build. EXHALE.x md5 f4da21f29579d87370efec2a4220f4c4, no warning, copied to LHS1140b/models/EXHALE_f4da21f2.x with its manifest. On LHS 1140 b the two He items move the He 10830 EW by +0.05 per cent (He/H 1.8), but a state of the previous physics does not certify under them, so the molecular states are re-solved on this binary.

The molecular K_zz = 1e9 family under the physics of section 22 (EXHALE_50aa18d7.x; .P1/newphys/, .P1/refine_grid_r3/). He/H 1.8 and 2.13 re-solved from the round-2 states; 1.6 and 1.52 as rungs from 1.8 (1.45 failed its rung; the H2 carrier row held at 1e-1 through four passes). MEASURED, every evaluation certified except the one marked:

He/H	EW, 500 cells	1000 cells	2000 cells	continuum (DERIVED)
1.52	1.0038			
1.6	1.0475	1.0187	1.0039	0.988
1.8	1.1468	1.1162	1.1003 (a)	1.083
2.13	1.2853	1.2516	1.2343	1.216
crossing of 1.108	1.721	1.783	1.819	1.860

(a) the re-evaluation of the certified state refuses the base mass row by a factor 1.9 (1.4e-8 against 7.4e-9); its EW is

quoted for reference.

The EW converges at first order with the grid (successive-difference ratios 1.93-1.95, $p = 0.95-0.97$), as under the physics before the audit, and the crossing moves from He/H 1.46 (500 cells) and 1.55 (continuum) before the audit to 1.72 and 1.86. The 60 R_p domain factor of section 14 was measured before the audit and is not applied here.

The documents brought to the code (md/user_manual_update_20260926.md): docs/EXHALE_user_manual.tex (rebuilt, 103 pages, no error or undefined reference, every changed page looked at), README.md and README_HOWTO.md now describe the input keys (H_rec_escape, He_rec_coupling, the metals.inp keys cx_N2p_H, cx_O2p_H, cx_full), the 24 columns of Heating_breakdown.txt (charge-exchange heat in column 24), the stage_channels.txt layout and the atomic data of sections 17-24. Found in the code and fixed: H_rec_escape was parsed but missing from the list of recognized keys (a spurious "unrecognized input line" warning, and a second statement not refused); the missing-metals.inp message promised built-in abundances that are zero; comments said cx_full toggles the He-H pair and that O2+/N2+ + H are dropped; the stage_channels.txt header omitted the metal charge-exchange channels 27-32. md/input_schema.md K14e (still "Koskinen 2013") and its metals.inp table (no group-E keys) were brought up to date. Not changed: H_rec_escape and He_rec_coupling, like the other physics switches of their group, are neither echoed in the setup report nor restart option tokens.

Checks of the binary of this section (EXHALE_f4da21f2.x). The suites: physics_probe did not compile one new probe (inner_shell_photoabsorption_ledger called cx_set_cell without the electron density the density-dependent reverses added; the two items were written at the same time); fixed, and the suite passes. grid_and_gates fails its fixture rows plus outer_iteration_ending_is_the_stagnation_one and restart_lower_ghost_is_the_derived_boundary (7.9e-14 against 1e-14), all on the pinned fixture of the previous physics; residual_determinism keeps its known row; the two steady_* suites need a solve log. The regression matrix ran all 17 cases; every one differs from the goldens by the physics (wasp_full log10 Mdot and the base inflow: 214 inflow cells against 257). **The user held the golden refresh (2026-09-26)**. The stationary re-solve of the fixture from the previous one did not converge (info = 2: the energy row 3.1e-2 at cell 406, the closure of System_HeH_TR_metals 4.6e-3 at cell 340), while the matrix case wasp_full_newton (marching, then Newton) certified (info = 0, ||R|| 4.4e-9); that output is the candidate fixture. The tree was rebuilt after the comment and message corrections of this section (EXHALE.x md5 0a7532d5087f14df41142fb499b68240; it differs from EXHALE_f4da21f2.x in printed strings and comments only).

The molecular states on EXHALE_f4da21f2.x (.P1/newphys/HeH<x>_r4, re-solved from the certified states of the previous physics; certified in 9, 9, 8 and 7 passes; every evaluation certified). He 10830 EW, 500 cells: He/H 1.52 1.0047, 1.6 1.0483, 1.8 1.1477, 2.13 1.2862 (+0.06-0.09 per cent against the table above); the 1.108 crossing 1.7194 (1.7213 before). The refined grids were not re-solved on this binary; the crossing's grid factor above stands within the size of this change.

25. The helium-hydrogen charge exchange from its originals (2026-09-26)

The He2+ + H original the user supplied (West, Lane & Cohen 1982) and an analysis made on the MoCHII side of the He+ + H exchange (~MoCHII/MoCHII_v1.00/md/heii_h_charge_exchange/) were checked against the published papers and applied (md/charge_exchange_detailed_balance_20260924.md section 12.9). Every rate below

is DERIVED from the cited cross sections unless marked READ.

- **He2+ + H -> He+(1s) + H+ + photon.** The $1.0\text{e-}14 \text{ cm}^3 \text{ s}^{-1}$ of Kingdon & Ferland (1996) is not a result of any calculation: it is the value they set for every reaction without a significant channel (READ, their sect. 2). Now the Maxwellian average of the West et al. (1982) radiative cross sections (Figs. 4-6, with the resolved resonances): $2.0\text{e-}13$ at $1\text{e}3 \text{ K}$, $1.65\text{e-}13$ at $1\text{e}4 \text{ K}$, $1.57\text{e-}13$ at $3\text{e}4 \text{ K}$, 16-20 times the old value; a calculation above $1\text{e}3 \text{ K}$, an extrapolation below.
- **He+ + H -> He + H+, non-radiative.** The Kingdon & Ferland fit reproduces the singlet-conditional cross sections of Zygelman et al. (1989) without the 1/4 of the spin statistics, and those cross sections are themselves hundreds of times those of Kimura et al. (1993) and Loreau et al. (2014). Now the Loreau et al. (2014) H+ + He cross section by microscopic reversibility (weights 4:1, threshold 13.65 eV/u, zero below 15 eV/u): $4.4\text{e-}17$ at $1\text{e}4 \text{ K}$ instead of $9.0\text{e-}15$, 1.5 per cent of the radiative channel; the band at the code (up to 2.9x at $1\text{e}4 \text{ K}$ from the threshold treatment; 0.4-2.5x above $3\text{e}4 \text{ K}$). The reverse He + H+ (B1) follows by detailed balance ($5.1\text{e-}22$ at $1\text{e}4 \text{ K}$ instead of $1.0\text{e-}19$); the heat of an event is unchanged.
- **He+ + H, radiative.** Stancil, Lepp & Dalgarno (1998) Table 1 row 19 prints $1.20\text{e-}15$ (READ); the code carried the $1.25\text{e-}15$ of Glover & Jappsen (2007). Now $1.20\text{e-}15$.

Effect (MEASURED): `wasp_full log10 Mdot -0.002` (inside the marching spread), He III down to 0.94 dex at the base, He II and He 2³S +48 and +52 per cent at 1.1 R_p, the charge-exchange share of the heating 5.6 -> 0.9 per cent at 1.05 R_p. **LHS 1140 b He/H 1.8 (evaluation of the certified state): the old He+ + H exchange supplied up to 10.7 per cent of the heating; without it the peak temperature falls from about 5600 to 5150 K, He II and He 2³S rise by 14 and 15 per cent at 1.5 R_p, and the He 10830 EW rises from 1.1477 to 1.1818 %A (+3.0 per cent).** The certificate does not hold under the corrected rates (energy row 8.4e-2), so the molecular states are re-solved on the binary of this section (EXHALE.x md5 59c6be36820b8ba5c958c294ad9e6f02, copied to LHS1140b/models/EXHALE_59c6be36.x with its manifest).

The molecular states re-solved on EXHALE_59c6be36.x (.P1/newphys/HeH<x>_r5, from the certified states of section 24): He/H 1.6, 1.8 and 2.13 certified in 25, 23 and 21 passes, every evaluation certified. He 10830 EW, 500 cells (MEASURED): 1.6 1.0577, 1.8 1.1550, 2.13 1.2902 (+0.9, +0.6 and +0.3 per cent against section 24; the +3.0 per cent of the evaluation at fixed wind is mostly given back by the solved wind); the 1.108 crossing moves from 1.7194 to 1.7026 (DERIVED). He/H 1.52 stopped in its second pass: a JFNK trial iterate with ||R|| near 2 left a lower ghost whose base H2 handoff partition did not close in 30 passes (last move $2.0\text{e-}6$ against $1\text{e-}10$), and the guard of `ionization_equilibrium.f90` (a boundary nothing stands behind is refused) ends the run with error stop even on a trial state. Whether a trial should instead be refused and the step shortened is a solver-control question put to the user.

26. A trial state no longer stops the run at an unclosed lower ghost (2026-09-26)

User decision: as recommended (`md/ghost_closure_trial_refusal_20260926.md`). The guard of `ionization_equilibrium.f90` that stops the run when the lower ghost's base H2 handoff partition does not close against its own ionization balance (tolerance $1\text{e-}10$, 30 passes) now distinguishes a state a caller can refuse from one it cannot: the stationary JFNK's trial evaluations (`eval_residual` of `steady_newton.f90` when it screens

admissibility: line search, trust region, Levenberg-Marquardt and pseudo-time trials, the finite-difference probes) and the relaxation's trial chemistry (equilibrate_chemistry_at_fixed_conserved_state) declare it before the sweep, and there an unclosed ghost refuses the trial (logged as CANDIDATE REFUSED, counted in the ledger, refusal code 7, closure reason 5), the solver then shortening the step as for its other refusals; an accepted state, the state written, the marching sweep and every other path keep the stop. The same rule now covers the ghost-composition fixed point one step later. MEASURED: the He/H 1.52 re-solve of section 25 passes the trial that stopped it (the $\lambda = 1$ trial of JFNK solve 3 iteration 7; the halved trials close) and certifies at pass 24; an evaluate of He/H 1.6 and a copy of mol_carrier are byte-identical to the build before (EXHALE_59c6be36.x) in every data line; the suites pass, the two failing ones (residual_determinism, grid_and_gates fixture rows) failing identically on the build before. EXHALE.x md5 7db5afe9b1d80ce757fcc07d1b499330, copied to LHS1140b/models/EXHALE_7db5afe9.x with its manifest.

The He/H 1.52 state on this physics: EW 1.0149 %A (certified evaluation); with 1.6, 1.8 and 2.13 of section 25 the 500-cell ladder is 1.0149, 1.0577, 1.1550, 1.2902 and the crossing of 1.108 stays at 1.7026.

The grid of the crossing on this physics (.P1/refine_grid_r5/, binary EXHALE_59c6be36.x, the 1000-cell He/H 1.8 and 2.13 continued on EXHALE_7db5afe9.x, identical physics; every state and evaluation certified). He 10830 EW, MEASURED; continuum DERIVED (Richardson at the measured order):

He/H	500 cells	1000 cells	2000 cells	continuum
1.6	1.0577	1.0297	1.0150	0.999
1.8	1.1550	1.1249	1.1094	1.093
2.13	1.2902	1.2571	1.2403	1.223
crossing of 1.108	1.703	1.764	1.797	1.838

First order again (successive-difference ratios 1.91-1.96, $p = 0.93-0.97$). Against the physics before the audit (1.464 on 500 cells, 1.552 in the continuum) the crossing has moved to 1.703 and 1.838; the continuum value is interpolated between the certified continuum points of 1.8 and 2.13.

27. The molecular family under the corrected physics: catalog publication and the width of the line (2026-09-26)

Catalog. The certified He/H 1.52, 1.6, 1.8 and 2.13 states of sections 25 and 26 were published into LHS1140b/models/molecular_scalar_gj1132_kzz1e9/ (run_case.sh, SEED= the certified run, EXHALE_7db5afe9.x): every re-measurement "original claim REPRODUCED", every evaluation certified; EW 1.0149, 1.0577, 1.1550, 1.2902 %A, $\log_{10} \dot{M}$ 7.89, 7.90, 7.91, 7.93. The catalog's He/H 0.55, 1.45 and 9.7 molecular states are of the physics before the audit and were not re-solved.

The added Gaussian (.P1/connectivity/broaden_molecular_final.py, the operations of section 13; output broaden_molecular_final.txt, figure docs/figures/lhs1140b_kzz1e9_molecular_broadened_final.pdf). The memo check (atomic well-mixed He/H 0.40) reproduces $f = 22.22$ km/s. MEASURED: the matched kernel is 22.10-22.19 km/s FWHM on every state and grid (σ 9.39-9.42 km/s), and at it the red depth, blue depth and

ratio of the He/H 1.8 state lie inside the observed 1 sigma (red 1.289, blue 0.170, ratio 7.60). The broadened EW (MEASURED on each grid, continuum DERIVED):

He/H	500 cells	1000 cells	2000 cells	continuum
1.6	1.0227	0.9955	0.9813	0.966
1.8	1.1167	1.0876	1.0726	1.057
2.13	1.2474	1.2155	1.1992	1.182
crossing of 1.108	1.781	1.851	1.890	1.933

As turbulence the kernel is supersonic: on the He/H 1.6 and 1.8 states the line-forming region ($n(2^3S)$ r weighting) has $T = 2650$ and 2756 K, $\mu = 1.94$ and 2.03 , $c_s = 4.33$ km/s, so $\sigma = 9.4$ km/s is Mach 2.2 as a line-of-sight dispersion, 3.1 as $\sqrt{2}$ sigma and 3.8 as an isotropic speed, as before the audit. From the EW alone the crossing is 1.703 (500 cells) and 1.838 (continuum); from the broadened line 1.781 and 1.933.

28. The catalog summary brought to the tree's state, and why the molecular winds do not certify (2026-09-26)

docs/lhs1140b_catalog_state.tex (13 pages, rebuilt, pages 5 to 8 checked by eye):

- The certified table carries the molecular $K_{zz} = 1e9$ rows the catalog now publishes, READ from each case's run.log and models/status.py: 1.52, 1.6, 1.8 and 2.13 under EXHALE_7db5afe9.x (note a); 1.45 under EXHALE_b059a4be.x, the second round of corrections, EW 0.9387 (note b; under EXHALE_50aa18d7.x neither its re-solve nor its rung from 1.52 certified within the time ceiling, and it was not attempted under the binary of record); 9.7 (note c, below). The rows of the table before this entry were the pre-audit values (1.6: EW 1.2155, now 1.0577).
- A new subsection of section 3, "Why the molecular winds do not certify", assembled from md/lhs1140b_stationary_L34c_20260918.md, md/lhs1140b_alteration_audit_20260921.md, md/lhs1140b_coupled_trial_20260921.md and md/TO_BE_DONE.md L22: the composition half of the partitioned route refuses (the H2 carrier row in all six cases), by two measured mechanisms (the composition update takes the wind out of solution in the well-mixed cases; a drift under a saturated movement bound in $kzz1e9/HeH0.083$); the coupled route fails in its Krylov solve; the open item is a preconditioner that holds the species rows. Every one of those measurements predates the atomic-data corrections and none was repeated under EXHALE_7db5afe9.x.
- The two molecular cases that published no state (molecular_scalar_gj1132_wellmixed/HeH2.13, molecular_photochem_gj1132_kzzprofile/HeH2.09) are now named.

Found and fixed: LHS1140b/models/status.py printed the not_solved.md reason beside molecular_scalar_gj1132_kzz1e9/HeH9.7, which is certified. The certified generation g0008_20260920T052941Z_c3c466a4 is the 2026-09-20 evaluation of the state written on 2026-09-19; the not_solved.md is that of a re-solve made later the same day (g0006_20260920T000506Z_5b2061aa,

composition_refusal, the H2 carrier row 5.13e-4 at cell 262 after 40 passes), which did not replace it. A certified row with a not_solved.md beside it now reads "a later re-solve, not the state shown: ..." (MEASURED on the table output). MODELS.md was not rewritten.

29. Errors fixed and models solved first: the plan of 2026-09-26

User instruction (2026-09-26): re-checking solutions that move little against the observational error is not the priority; fixing errors and solving the models that do not solve is. Plan md/PLAN_20260926.md; the user approved its recommendations (goldens and fixture refreshed; the composition-side solver work reopened if continuation does not close the cases; diffusion_check taken out of the model count). New physics only where it can change results seriously; the rest is recorded, not applied.

29.1 An OpenMP data race in the metal sweep (fixed)

meg_g02 (the inner-shell Auger share of neutral photoionization to X++, added with binary f4da21f2 on 2026-09-25) was missing from the private list of the He + metals cell sweep (ionization_equilibrium.f90, then ~2137, a default(shared) loop), so threads overwrote each other's value between the write and the two reads. MEASURED: route-B stationary restart of wasp_full_newton failed at 8 threads (composition elimination never contracts, info = 2) and certified at 1 thread; two 8-thread runs of one input differed before the solve started (md/metal_sweep_openmp_race_20260926.md). Every metals-on run at more than one thread since f4da21f2 was affected; the regression matrix (1 thread) and every metals-on catalog certificate (last published 2026-09-20) were not. Fix: meg_g02 private, and all four cell loops of ioniz_eq converted to default(none) with explicit shared lists; an audit of every parallel region of radiation/ and lower_atmosphere/ found one more, formal, race (two table-ready flags read without synchronization; now omp atomic read/write seq_cst). New suite src/tests/thread_count_identity/ (200 steps of wasp_full at 1 and 8 threads, data lines bitwise): 7db5afe9 FAIL (3.2e-3 Hydro, 5.4e-2 Ion), fixed binary PASS. The r0 case (route B, 8 threads) now certifies, info = 0, ||R|| 6.50e-9. No 1-thread run moved.

29.2 Fix batch (errors confirmed in the source)

F1 free-free Gaunt factor: van Hoof et al. (2014) eq. (21) uses the infinite-mass Rydberg (READ, their sect. 2), so $Ry/k = 157887.51$ K, not 157807 (the Hui & Gnedin 1997 constant, kept in that fit); F2 the carrier diffusivity read by outputs and diagnostics was zero on a stationary path (MEASURED $D_{H2} = 0$ at every cell; now $7.9e6 - 8.2e9$ cm²/s); F3 the determinism replay now passes u to cell_row_scales; F4 pressure formed before the composition refresh at 3 of 9 sites (coupled-block entry, after solve_steady_*, trial end; the other 6 verified not stale); F5 the CO column, dissociation rate and shielding factor joined the step checkpoint; F6 an inert Coupled carrier solve is reported; F7 five wrong comments; F8 the _adv headers carry every provenance field; F10 seven callerless Cool_coeff wrappers removed and the Voronov H/He fit defined once (rates bitwise unchanged). Effect on the regression inputs (single thread): F1 alone, at most 1.4e-4 relative (wasp_full velocity), log10 Mdot unchanged to 4 digits.

29.3 The interdiffusion enthalpy flux (new physics, serious)

The energy equation of a diffusing mixture carries $q_d = \sum_i h_i J_i$ (Cook 2009, Phys. Fluids 21, 055109, eqs. 11-13; Schunk 1977, Rev. Geophys. Space Phys. 15, 429, eqs. 15b, 16, 20c; READ). EXHALE moved He relative to H and carried no such term in either energy equation. MEASURED size on LHS 1140 b He_diffusion states: 13-26 % of the local energy budget at 1.1-5 R_p (md/interdiffusion_enthalpy_flux_20260926.md). Implemented in both routes (binary model: $q_d = J_{He} (h_{He} - h_H)$, sensible enthalpies with the electrons each ion carries; formation energy verified to be carried already); key Interdiffusion enthalpy flux: default True with He_diffusion (the term belongs to the equation), False only to match published models that omit it (Koskinen 2013, 2022; Yelle 2004); restart token interdiff_enth (older states restart under Restart option change: interdiff_enth). Suite src/tests/interdiffusion_enthalpy_flux/ 18/18. Effect on the molecular He/H 1.8 state (24 + 24 passes, carrier row 1.6e-5 at the end, not yet certified): T -166 K median over 1.3-2 R_p (largest -236 K, -4.4 %), He/H -3.1 %, the $n(2^3S)$ column -1.3 %, $\log_{10} \dot{M}$ +0.002.

29.4 He+ + H non-radiative charge exchange near threshold

Section 25 took the Loreau et al. (2014) H+ + He cross section forward by reciprocity with $\sigma_r = 0$ below 15 eV/u, where their calculation starts. That closes an open channel: the forward (exothermic) rate then fell as $\exp(-E/kT)$ at low T (5e-34 at 300 K). Adopted, as in MoCHII (the user's decision relayed by the MoCHII session), $\sigma_r(x) = \sigma_r(15) \sqrt{(x - x_{th})/(15 - x_{th})}$ between $x_{th} = 13.655$ eV/u and 15 eV/u, i.e. σ_f v constant at low energy (ion-induced-dipole capture with a constant transfer probability; an assumption no calculation tests). Rate: 9.64e-17 at 1e4 K (was 4.41e-17), 9.56e-17 at 1e3 K (was 1.5e-21); 3.3 % of the radiative channel at 1e4 K. Generator cooling_data/heii_h_charge_exchange_rate.py (reproduces the old table to 0.4 % with the old cut).

The radiative channel, the user asked, is one coefficient in both codes: Stancil, Lepp & Dalgarno (1998) Table 1 row (19), $1.20e-15 (T/300)^{0.25}$, the Zygelman et al. (1989) Table I DIRECT column times the 1/4 of the singlet approach (READ; 1.17e-15 at 300 K from the original). MoCHII moved from the Glover & Jappsen (2007) R26 1.25e-15, which cites the same data and cannot be reproduced from it (2026-09-26); EXHALE unchanged. A digitization of Courtney et al. (2021) Fig. 3 agrees with this fit within 2 % over 1e3 - 1e4 K and lies 1.9x above it at 100 K; a fit to their tabulated rate needs the data file (University of Georgia Molecular Opacity Project, not reachable from here).

29.5 Goldens, fixture, suites

Goldens re-snapshotted from the tree binary 4e275f1d (the physics of 7db5afe9) after a full matrix (all 17 cases differed from the round-3 goldens, as expected); backup golden_pre_round45_20260926. Fixture wasp_full_newton/IC replaced by the 1-thread route-B certified state of 7db5afe9 (backup IC_pre_round45_20260926). Every suite on the integrated binary: all pass except grid_and_gates (3 rows: the fixture is not a root of the fixed physics, to be re-made with the final binary), residual_determinism (known row), the two suites that need run logs.

Closing the series (2026-09-27): the matrix on the final binary 6844f188 (sections 29.1-29.4, 29.7, 29.9) differs from the goldens of 7db5afe9 in every case beyond 1e-3, largest in the helium columns of the lower cells (3.7 % at row 1 of wasp_full, the He+ + H rate at low T) and in the molecular cases (the ghost start, the progress control); gold-

ens refreshed from it (backup golden_pre_final_20260927). Fixture re-solved on 6844f188 at 8 threads (info = 0, lIRll 4.11e-9; backup IC_pre_final_20260927); grid_and_gates 345/345. Every suite passes except the known residual_determinism row and the two suites that need run logs.

29.6 Models solved

MEASURED, .P1/s1_remeasure/, .P1/s2_continuation/: every failure of the catalog's unsolved molecular and atomic cases stands under 7db5afe9 (six passes each); continuation then solved:

- molecular_scalar_gj1132_kzz1e9/HeH1.45: rung from the certified 1.52, certified in 13 passes.
- molecular_scalar_gj1132_wellmixed/HeH2.13, 1.8, 1.4: the well-mixed molecular cases differ from the $K_{zz} = 1e9$ ones ONLY by He_diffusion off (their base.inp carries Kzz_base 1.0e9, which the carrier transport reads). From the certified kzz1e9/2.13 state, He/H set uniform (new map_state_to_grid.py --uniform) and diffusion off: certified in 10 passes; then rungs 1.8 (10 passes) and 1.4 (14 passes) with make_rung.sh (UNIFORM=1). The direct re-solve of the catalog's well-mixed 2.13 state is refused by the progress control at pass 5, so the continuation is what solves it.
- molecular_photochem_gj1132_kzzprofile/HeH9: on the final binary of the series (2dd9a865, face at the level), mapped from its 40-pass state on 6844f188 (.P1/s2_continuation/photo9_f, row 7.4e-5 falling 0.77 a pass) and continued 40 passes: certified (H2 carrier row 9.2e-6 at cell 217), .P1/s2_continuation/photo9_v4 (2026-09-27). Its earlier stops were the lower-ghost closure defect of section 29.9.
- Ionization-stage transport on the molecular 1.8 state stalls on the He++ row under the progress control; the cause is the control (29.7).

29.7 A progress-control defect of the composition relaxation

Diagnosed (.P1/iontr_diag/): each pass of the carrier relaxation ends on its particle-count cap, the state moves a fixed amount a pass, the relative gated measure stays flat because residual and scale fall together, and the outer control reads that as no progress, halves the cap and ends the loop. With the cap held (diagnostic override) the same state certified in 14 passes.

Fixed (EXHALE_main.f90 progress control, binary d4b571e8): a pass whose consumed carrier update ended on its movement bound, whose solve converged and whose joint distance did not rise by more than 1.1x is not judged by the relative measures and does not halve the bound; the bound doubles after such a pass up to 0.5 and halves when a bound-limited update raises the distance of a converged wind or precedes the first failed solve; omega follows the same rule and no longer drops on hydro entry failures; EXHALE_CARRIER_TRUST_HOLD still freezes the bound. Also fixed: the scale x_ref of the carrier fixed-point test was the H2 maximum alone, so with transported stages and no molecules it sat at its 1e-30 floor and the test could never pass (logged displacement ~1.7e28, MEASURED on iontrans_metals). MEASURED: the hot-Uranus stagnation fixture certifies at pass 42 (it stagnated; the suite row that needs that ending now holds the bound); iontrans_metals accepts at pass 17 instead of 48, the two certified states within 7.6e-5; the molecular He/H 1.8 state with transported stages certifies in 14 passes from a starting bound of 0.5 (EXHALE_CARRIER_TRUST=0.5) but not from the default 1e-2, where the hydro solve fails at pass 2 under the old

control and the new one alike; with the interdiffusion flux on, that state's hydro solve fails on entry and a slow mode on the He front (0.86-0.93 a pass) remains. REGRESSION on kzz1e9/HeH0.083: a growing bound makes the H2 row at the outer cells 498-500 overshoot (up to $1.5e-1$) and the bound saws; neither control certifies that case (the old one stalls on the elemental row at cell 338). mol_carrier bitwise unchanged (the outer loop is not entered). Open: the starting bound for transported stages, the H2 outer-cell overshoot, the hydro entry failures with the interdiffusion flux.

29.8 Incidents of this series

- A sed insertion with an empty line address put a three-line comment before every line of src/tests/boundary_state/molecular_seed_of_an_atomic_state.sh; the 153 blocks were removed and the file checked by diff against the afternoon copy (identical except the intended line).
- The tree binary was rebuilt while a regression matrix was running on it; the matrix was stopped and is re-run after the series.

29.9 The lower-ghost H2 closure stalled with a zero move (fixed)

The molecular photochemical-column cases (He/H 9 and 2.09) stopped at passes 1-3 with "the base handoff partition of a lower ghost did not close": the closure's own move exactly zero and its residual 9-41 times the $1e-10$ tolerance. Diagnosed (.P1/ghost_stop/): each closure pass after the first started the molecular solve from the previous pass's accepted state, whose x(4) held the OLD imposed H2 partition; the pinned row x(4) - x_h2_fix then entered hybrd1 carrying the whole secant move, and hybrd1 (xtol = sqrt(eps), diag = 1) can return a move below $2 \text{ xtol } ||x|| \sim 3e-8$ untouched with info = 1 (hybrd.f90: first delta = pnorm, halved on the rejected step, then delta <= xtol xnorm). The write-back then imposed the value, the next pass's map saw no move, and 28 identical solves followed (27 of 27 traced events; 41 of 41 refusals and stops in the S1 logs, residuals $8.2e-10$ - $2.8e-8$, all below that floor). Fix: impose_transportated_fractions sets every unknown a transport solve owns (H2 partition, oxygen carriers, ionization stages) to its imposed value before both the molecular and the atomic hybrd1 solves; the stop report prints the pin error. MEASURED with the fix (worker build): 0 of 127092 ghost closures unclosed, pin error exactly 0, no refusal and no stop; the He/H 9 case then fails on its H2 carrier row (0.88), a separate problem. Binary 6844f188. Every molecular base-handoff regression case is reached (ghost composition moves by $1e-12$ - $1e-9$).

29.10 The _adv post-process made second order

User decision (2026-09-27): the catalog quotes the He 10830 EW of the _adv post-process. That post-process was a first-order upwind recursion in r (backward Euler of $dx/dr = R/v$ with the rates at r_j and the residence time at r_{j-1}), the main source of the first-order grid convergence of the EW (md/ew_grid_order_20260926.md). Now the variable-step BDF2, $y_j - a_1 y_{j-1} + a_2 y_{j-2} = g h_j F(y_j, r_j)$ with $w = h_j/h_{j-1}$, $a_1 = (1+w)^2/(1+2w)$, $a_2 = w^2/(1+2w)$, $g = (1+w)/(1+2w)$ (derived at post_process_adv.f90, coefficients as eq. (1.4) of Li & Liao, arXiv:2201.00527; L-stable, so a population that equilibrates within a cell does not ring as under the trapezoid rule), for every composition system and the energy equation, rates and $1/v$ at the same point; backward Euler on the first step, after a refused cell, for $w > 1 + \sqrt{2}$, and as a retake of a step with a negative population or no temperature root

(0 retakes on the nine states). MEASURED on the nine certified grid states (He/H 1.6, 1.8, 2.13 on 500, 1000, 2000 cells), with the state held fixed: the post-process share of the EW error falls from +3.0 % / +1.0 % ($p = 1.00$) to below $3e-4$, and the $_adv$ 2^3S column converges at $p = 1.98 - 2.14$ ($1.00 - 1.01$ before); a closed-form column test gives $p = 1.989$. The EW itself changes by -3.9 % (500 cells) to -0.97 % (2000 cells) at He/H 1.6 ($1.05692 \rightarrow 1.01571$ on 500 cells) and still converges at first order: what remains is the solved state's first-order base closure (being fixed) and the transit integration ($p \sim 1.4$, opposite sign).

Also fixed: the $h \operatorname{div}(\rho v)$ term of the $_adv$ energy equation was built from differences of the cell-centred $\rho v r^2$, which carries the odd-even pattern of the cell-centred velocity at the base ($1.8e12$ to $3.4e12$ between neighbours while the face mass flux changed by $1e-10$), giving 805 / 4008 / 579 K at the first corrected cell on 500 / 1000 / 2000 cells; it is now the state's own mass row, the face-flux divergence. Below about $1.13 R_p$ the molecule-free post-process lacks the molecular heating and $H3+$ cooling that set the run's energy balance there; by the user's decision (2026-09-27) the run T is kept there with status 3 "unsupported" wherever the omitted heating and cooling exceed the carried ones (196 / 389 / 776 cells on 500 / 1000 / 2000 cells, up to $1.134 - 1.137 R_p$; EW change at most $1.2e-4$ relative); the user manual's status table says so. Binary ca651768. Suites: $_adv_static_limit$ 53, $_transit_state$ 12, $_transit_census$ 21, $_physics_probe$ 2199, $_thread_count_identity$ 9, all pass. Every $_adv$ golden moves (5 short cases: T up to 5.9 %, $n(2^3S)$ up to 9.4 %). The catalog EWs predate this change by -1 % to -4 %. Found, not changed: $_thermal_damkohler_number$ is the net rate $IH - Cl/u$, near zero at a balanced cell, not the relaxation rate its comment describes. Docs: the transit tool now says a stationary SOLVE writes no $_adv$ products and its evaluation does; the project CLAUDE.md says the matrix compares the $_adv$ pair as well.

29.11 The lower face at the reservoir level

The reservoir (the `base.inp` handoff or the Photochem profile's matching level) is stated at $r = 1$, but the domain's lower face sat at $r_edg(0) = 1 + dr/2$, and `base_boundary.f90` carried the reservoir pressure up to that face along the reservoir's own isentrope: a half-cell slab of 226 K gas whose thickness scales with dr lay between the level and the domain (MEASURED: the lower layer offset by -2.2 % / -1.15 % in density on 500 / 1000 cells, `md/ew_grid_order_20260926.md` section 3). Now the face is the level in every grid mode (Uniform, Stretched, Mixed: `define_grid.f90`, $r(0) = 2 - r(1)$, refusals if a ghost reaches $r \leq 0$ or the face is not 1), the face state is the stated reservoir with no transport along any isentrope, the ghosts hold the reservoir continued below the level, and the Photochem matching level is the face on every grid; boundary model id `..._face_at_level_v4`. MEASURED on the molecular He/H 1.8 state, 500 / 1000 / 2000 cells, all certified: the pressure gap between the interior at the level and the reservoir falls from 2.93 / 1.44 / 0.70 % to 0.114 / 0.067 / 0.038 %; the lower-layer density offset falls 20-40x; the wind quantities stay first order with errors 1.6-1.9x smaller ($\log_{10} \dot{M}$ 7.9147 / 7.9113 / 7.9094 $\rightarrow$ 7.9116 / 7.9098 / 7.9086; $_adv$ EW 1.1548 / 1.1248 / 1.1091 $\rightarrow$ 1.1482 / 1.1214 / 1.1074 %A, measured before the $_adv$ change of 29.10). The remaining first-order error is likely the thermal layer at the contact, which the grid does not resolve (cell-1 T 800 / 695 / 604 K against the 226 K reservoir, 13 % lower at every halving; not shown to drive the wind). Two other first-order terms were found and left: the element-diffusion Dirichlet He/H is imposed at the cell-1 center, $dr/2$ above the level (`binary_element_diffusion.f90` 946-959; the He/H change near the base is $1.5e-4$), and the conduction bath at the ghost center $r(0)$ (`viscous_conduction.f90` 485; conduction off in these runs). Every cell

center moves by $dr/2$, so the grid guard refuses every state written before: such a state loads only as a seed made with `map_state_to_grid.py --ic` onto the new grid (a 1-step cold run gives the target grid). The regression fixtures and goldens are migrated in the same series (MEASURED, 2026-09-27: every fixture of the default matrix and every fixture a suite reads mapped with `map_state_to_grid.py --ic` onto a 1-step cold-run grid; `wasp_full_newton/IC` re-solved and certified, `info = 0`, `||R||` 1.43e-9, and `iontrans_metals/IC` a copy of it; the `hp_*` seeds, whose recipe from `mol_base_handoff` is now refused by the carrier restart token, mapped from their previous pairs with the seeds kept (`x(H II)` 0 and 1e-12); `jfnk_hd189`, a named case outside the matrix whose legacy pair has no # columns header, not migrated; the two suites that read catalog states now map them into their own scratch; every suite passes, and the standing `residual_determinism` row now passes (5.9e-2 against < 1); goldens refreshed from the matrix run on 2dd9a865, backup `golden_pre_facelevel_20260927`; the LHS 1140 b catalog states stay records of the previous boundary (their EWs move by -0.6 % at 500 cells, below the observational error). Binary 2dd9a865.

29.12 A restartable state after every outer pass

The stationary route wrote its state only when the run ended, so a run stopped from outside (a wall-clock timeout) lost every pass it had made: MEASURED on 2026-09-28, three LHS 1140 b continuations lost 35, 13 and 9 passes. Now each outer pass that does not end the iteration writes the state the next pass starts from, `output/Hydro_ioniz_last_pass.txt` and `output/Ion_species_last_pass.txt`, overwritten each pass, by the same writer and headers as the final state files, rebuilt from (u, f_{sp}) in the order of `rebuild_state_from_checkpoint` with every side effect on the caloric mixture undone; both files carry `certified=F` `cert_reason=pass_snapshot_p<n>` and a # `pass_snapshot` line (the movement bound and omega of that pass, which a restart does not inherit: pass them by `EXHALE_CARRIER_TRUST / EXHALE_DIFF_OMEGA`), so a pair mixed from two passes is refused by `load_IC` and no tool reads a snapshot as certified. Each file is written to `<name>.part` and renamed. MEASURED: 0.03 - 0.05 s a pass on 500 cells (passes of 90 - 135 s); a run killed during pass 3 left the complete pass-2 pair, which restarted and continued as the uninterrupted run did; the final state files of a normal run are bitwise those of the previous binary. LHS1140b/models/.P1/s2_continuation/config_rung.sh continues from the pair when the parent has no final state. Binary a7255a03. The regression matrix on it, strict (`REGRESSION_REL_TOL=0`): all 17 cases byte-identical to the goldens of 2dd9a865 (MEASURED).

29.13 The unsolved molecular cases by continuation (state of 2026-09-28 16:30)

All runs under LHS1140b/models/.P1/s2_continuation/ (scripts `config_rung.sh`, `make_rung.sh` in .P1/, `chain.sh`, `facelevel_continue.sh`; env as the catalog: `EXHALE_PTC_DTAU0=1.0`, 8 threads). Certified states (MEASURED, `certified=T`, the JFNK `||R||` of the last solve 6.5e-9 - 2.9e-8):

family	He/H	passes	binary	note
kzz1e9	1.45	14	7db5afe9	rung from 1.52
kzz1e9	1.30	5 (after 24)	7db5afe9	
kzz1e9	1.223	9 (after 40 + 40)	2dd9a865	moved to the face-at-level grid

family	He/H	passes	binary	note
kzz1e9	1.186	24 (after 30)	2dd9a865	
kzz1e9	1.168	20	2dd9a865	
kzz1e9	1.15	46	a7255a03	carrier bound held at 1e-2
wellmixed	2.13, 1.8, 1.4, 1.1, 0.85, 0.7603, 0.68, 0.6116	10 - 21 each	7db5afe9	path finding, previous grid
wellmixed	0.6116, 0.55, 0.4	16, 18, 17	2dd9a865	on the face-at-level grid
photochem kzzprofile	9	41	2dd9a865	after 40 passes on 6844f188

Not solved yet: kzz1e9 below 1.15 (1.12 running, gated row falling to 1e-4 then flat at 1.3e-4 over passes 14-16); wellmixed 0.3347 and below; photochem kzzprofile 2.09 (gated row oscillating 0.08 - 0.54, hydro info = 2 on most passes, about 40 min a pass); kzz1e9 0.083 and the wellmixed 0.083 are the far ends of the two chains.

What was learned on the way (MEASURED):

- The well-mixed molecular cases are the kzz1e9 ones with He_diffusion off (their base.inp carries Kzz_base 1.0e9, which the carrier transport reads); a kzz1e9 state with He/H set uniform (map_state_to_grid.py --uniform) certifies with diffusion off in 10 passes, which is how the family was entered.
- Runs stopped by a wall-clock timeout lost every pass (35, 13 and 9 passes), which section 29.12 fixed for every later run; and config_rung.sh ran under set -e, so a run that timed out never wrote its EXIT.txt and a waiting continuation never started (fixed).
- The growing carrier bound of section 29.7 drives a sawtooth in the H2 cases below He/H ~1.2 (kzz1e9 1.15: the gated row between 5e-3 and 2e-1 over 35 passes with hydro info alternating 0/2) and in the well-mixed 0.3347 rung (between 2e-4 and 1e-2, the refusing cell moving between the front near cell 357 and the outer cell 500). Holding the bound at 1e-2 (EXHALE_CARRIER_TRUST_HOLD=1e-2) certified kzz1e9 1.15 in 46 passes, and 1.12 then fell 2.6x a pass to 1e-4; it did not stop the well-mixed 0.3347 oscillation (1e-3 - 2.7e-3 over 19 passes). The rule that grows the bound is therefore not right for the H2 carrier; an open item.

Documents brought to the code of this series (2026-09-28): the user manual (input keys Interdiffusion enthalpy flux, Outer shells, Composition update holds, H2 double ionization, H2 neutral dissociation and the retired Oxygen transport added from known_keys; the _last_pass output files; the face at the level; the BDF2 post-process; the progress control; the thread-count note; stale statements corrected, among them the tokens a Restart option change: line may not name, four not six), the physics and algorithms document (the interdiffusion enthalpy flux in the energy equations, the lower face, the grid constructions, the He+ + H pair, Ry/k, the post-process step, the progress control and the snapshot; stale statements corrected, among them the default of H2 double ionization, chung80, and the claim that every extension defaults to off) and README.md. Two source

comments corrected (base_boundary.f90 model id, steady_residual.f90 energy row scale; comments only), and src/tests/interdiffusion_enthalpy_flux/run.sh made executable: make test had skipped it silently.

Code-size summary: EXHALE vs. the original ATES (2026-06-07; updated 2026-09-22)

A line-by-line comparison of the Fortran source against the pristine upstream code (ATES/ATES-Code-main) quantifies the extent of the in-house extension. *Pure code lines (SLOC)* exclude blank lines and comments (the comment stripper is string-aware, so a ! inside a quoted string is not treated as a comment); the line-by-line diff additionally normalizes whitespace and case, so reformatting and comment-only edits are *not* counted as changes. The measurement script is src/utils/codesize.py, which reprints every number of this appendix and the byte-identical list of Table 11. Measured 2026-09-22, i.e. after the whole of stage 1 and of the stage-2 series through section 16 of the stage 2 log (md/Update_EXHALE_stage2.md): the acceptance and certification contracts, the attempted-step controller, the partitioned stationary solver, the oxygen and carbon chemistry, the molecular infrared bands, the ionization-stage transport and the charge-exchange rows, the base contact boundary and its trace, the molecular seed taken from an atomic state, the transition into the coupled carrier solve, the discrete conservation ledger, and the test suites those series brought with them.

Not yet re-measured for stage 3. The numbers of this appendix are those of 2026-09-22; the code changes of stage 3 (among them the atomic-data audit and its repairs, sections 17 to 19 of the stage 3 log) are not in them. They are re-measured with src/utils/codesize.py when that series is closed.

What the script counts as *core*. Every .f90 file under src/ except those in src/modules/wind_ae/, which is the ported Wind-AE IC generator and is listed separately. That rule puts the standalone drivers under src/tests/ in the core figure, and they are now the larger part of it by file count, so the table splits the core row into the simulation build and those drivers. Of the 102 files outside src/tests/, 101 are the SRC list of the Makefile and are compiled into EXHALE.x; the one exception is nonlinear_system_solver/constrained_equilibrium_probe.f90, which is built on its own. Each test suite likewise builds its own driver from src/tests/<suite>/run.sh.

	Fortran files	code lines (SLOC)
Original ATES (ATES-Code-main)	47	3 514
EXHALE core (excl. Wind-AE port)	208	93 928
of which the simulation build	102	61 129 ($\approx 17.4\times$)
of which the src/tests/ drivers	106	32 799
net change (core)	+161	+90 414 ($\approx 26.7\times$)
Wind-AE IC-generator port (wind_ae/)	+36	+4 756
EXHALE total	244	98 684 ($\approx 28.1\times$)

Line-by-line diff, core (comments + blanks excluded; whitespace/case normalized): **91 633** code lines added in EXHALE (new modules + the new side of modified lines) and **1 219** removed from the original (deleted + the old side

of modified lines), for **92852** total changed lines. At the file level, of the 47 original files **38 were modified**, **8 are unchanged** (all eight byte-for-byte; the list is Table 11) and **one was removed**: `flux/speed_estimate_HLLC.f90`, Toro's adaptive p_* estimate, was used by `Num_Fluxes.f90` and never called, because the HLLC branch builds its own Davis/Einfeldt speeds inline. **162 new core modules** were added. Of those, 56 are part of the simulation build:

New module	Purpose	SLOC
<code>time_step/steady_newton.f90</code>	stationary solve of the hydrodynamic rows: residual $F(Y) = 0$ on the physical cells, Newton/Krylov with pseudo-transient continuation, line search, best-iterate ledger	10796
<code>lower_atmosphere/diffusive_photochemistry.f90</code>	vertical transport of the molecular carriers (H_2 , OH, H_2O , CO) and of the transported ionization stages (H^+ , He II, He III), solved together with the chemistry that makes and destroys them	4152
<code>lower_atmosphere/molecular_infrared_data.f90</code>	band and line data of the molecular infrared coolant (generated by <code>cooling_data/molecular_infrared_bands.py</code>)	2182
<code>functions/binary_element_diffusion.f90</code>	conservative binary H/He element transport (stage-resolved pair coefficients, donor-cell drift, metals slaved to H)	2173
<code>time_step/certification.f90</code>	one evaluator of the equations a run actually solves, and the contexts that read its verdict	1696
<code>time_step/attempted_step.f90</code>	the attempted-step controller: one checkpoint, one trial, one adoption boundary, and the rollback contract	1608
<code>nonlinear_system_solver/constrained_chemical_equilibrium.f90</code>	chemical equilibrium of the molecular H/He network in logarithmic number densities, element conservation as explicit residual rows	1209
<code>lower_atmosphere/h2_self_shielding_table.f90</code>	H_2 Lyman–Werner dissociation cross section with overlapping lines, and its two branching ratios	952
<code>states/base_boundary.f90</code>	the lower boundary as a characteristic condition at the face, the number of conditions set by the Mach number there	598
<code>radiation/charge_exchange.f90</code>	charge-exchange network (Huang et al. 2023, Table 4), as residual rows and as the stoichiometric sources of the transported stages	560
<code>time_step/energy_semi_implicit.f90</code>	implicit source step for the thermal energy of a cell, at fixed composition and heating	535
<code>time_step/viscous_conduction.f90</code>	Navier–Stokes viscous momentum force, its dissipation, and thermal conduction (Crank–Nicolson)	529
<code>time_step/steady_residual.f90</code>	finite-volume stationary residual $R = du/dt$ and its norms, shared by the diagnostic, the convergence monitor and the stationary solver	528
<code>init/molecular_seed_from_atomic_state.f90</code>	the molecular state a solved atomic state of the same wind implies, written as an initialization file: an imposed H_2 partition capped by each cell's own neutral hydrogen, closed on the caloric equation of state at fixed pressure or at fixed temperature	516
<code>radiation/excited_hydrogen.f90</code>	non-LTE $H(n=2)$ 2s/2p equilibrium (Christie, Arras & Li 2013) and the Balmer feedback	502
<code>lower_atmosphere/element_inventory.f90</code>	one stoichiometric map of the H, He, O and C nuclei of a cell, and the three constraints a composition state is judged by	448
<code>nonlinear_system_solver/System_HeH_mol.f90</code>	coupled H/He + molecular ($H_2/H_2^+/H_3^+/HeH^+$) system, turnover-scaled rows, with the gross production and loss of each row on request	439
<code>lower_atmosphere/molecular_reaction_heat.f90</code>	net chemical heat of the collisional reactions of the H_2 /He network, with the dissociative recombinations that leave a product excited charged to that level and not to the gas	416

New module	Purpose	SLOC
files_IO/lower_atmosphere_profile.f90	lower-atmosphere profile reader: log- p interpolation, elemental reservoirs, $K_{zz}(p)$	367
nonlinear_system_solver/ion_residual_core.f90	shared ionization-residual rows and analytic-Jacobian pieces of the <code>System_*</code> family	361
states/caloric_eos.f90	caloric equation of state: internal energy density, pressure and temperature, with the H_2 level ladder	340
lower_atmosphere/oxygen_rates.f90	rate coefficients and thermodynamic data of the oxygen-hydrogen chemistry, with their validity ranges	316
functions/element_census.f90	one definition of the elemental and charge content of a composition state, and the invariant checks built on it	307
time_step/coupled_block_handover.f90	the state at which the wind-composition alternation stops on its movement bound, and the one recorded pair (u, f_{sp}) the coupled carrier solve is entered from	298
radiation/electron_energy_degradation.f90	degradation of a fast photoelectron in a partly ionized H/ H_2 /He gas: the heat fraction and the ionizations of each target	287
radiation/lya_rt.f90	$Ly\alpha$ transfer by the escape-probability closure, evaluated in line each step	257
time_step/conservation_budget.f90	every term of the three finite-volume conservation rows of one assembly, with the geometry, the potential and the gravitational work the energy flux difference carries inside it	252
functions/ionization_stage_transport.f90	transport of the ionization stages of one element on that element's own nucleus flux, the neutral stage closing the simplex, so that moving charge between stages moves no nucleus	235
lower_atmosphere/molecular_infrared_cooling.f90	infrared exchange of the molecular layer: H_2O and CO vibration-rotation bands, H_2 quadrupole and magnetic-dipole lines	231
lower_atmosphere/h3p_cooling.f90	optically thin H_3^+ infrared cooling (Miller et al. 2013)	229
lower_atmosphere/mol_rates.f90	molecular reaction rates (Koskinen et al. 2022, Table 1)	216
functions/h2_photo_channels.f90	H_2 photoabsorption resolved into mutually exclusive final-state channels, with one stoichiometric table	199
lower_atmosphere/lyman_werner.f90	H_2 photodissociation in the Lyman and Werner bands (the Solomon process)	195
radiation/opacity_models.f90	opacity-model dispatcher for the photoionization cross sections (A/C/P/T)	180
files_IO/metals_input_read.f90	parse <code>metals.inp</code> : trace-metal abundances at runtime	177
states/boundary_state_trace.f90	the distance between the cached base face state and the one the installed composition would give now, measured and never decided on	172
flux/species_face_flux.f90	advection of the species mass fractions on the same faces, with the same mass flux, as the density	163
functions/composition.f90	single point turning (ρ, f_{sp}) into every species density, n_e and n_{tot} , and converting pressure and temperature	162
time_step/hydrodynamic_rows.f90	the hydrodynamic rows of the stationary residual in double precision, instantiated from the kind-generic text	161
lower_atmosphere/h2_vibrational_relaxation.f90	the fate and the size of the vibrational energy left in an H_2 molecule, for both of its drivers	155
nonlinear_system_solver/System_HeH_metals.f90	coupled H/He+metals ionization (+ analytic Jacobian)	153
nonlinear_system_solver/System_HeH_mol_metals.f90	molecular + metals merged ionization system	143

New module	Purpose	SLOC
init/species_table.f90	metadata table of the metal stages and of the base, molecular and oxygen species	128
lower_atmosphere/co_photodissociation.f90	CO photodissociation on the 912–1201 Å band	127
nonlinear_system_solver/newton_solver.f90	damped Newton with an analytic Jacobian for the ionization systems, with the MINPACK hybrd1 retry	127
lower_atmosphere/water_photolysis.f90	H ₂ O and OH photodissociation in the four far-ultraviolet bands of the oxygen option	111
lower_atmosphere/co_self_shielding_table.f90	CO self-shielding function $\Theta[N(\text{CO}), N(\text{H}_2)]$ (Visser, van Dishoeck & Black 2009)	110
states/stationary_operator.f90	the spatial operator a stationary state is a state of, and the face mass flux $r^2(\rho v)_f$ it puts through every face	107
nonlinear_system_solver/ion_cell_state.f90	named-field cell state of the ionization residuals, in place of the position-packed params	98
flux/low_mach_dissipation.f90	gated fourth-difference stress damping the two-cell velocity mode a contact-resolving flux leaves undamped	87
lower_atmosphere/lower_column.f90	analytic lower-atmosphere column from the 1-bar level to the base pressure (Koskinen et al. 2022)	87
files_IO/opacity_input_read.f90	parse opacity.inp	85
radiation/hydrogen_n2_rates.f90	atomic data and rate coefficients of the H $n=2$ manifold and of the Ly α line, in one place	76
init/blas_thread_policy.f90	the thread count of the LAPACK library the binary links, resolved at run time and set to one unless the run states it	73
nonlinear_system_solver/System_HeH_TR_metals.f90	merged He 2 ³ S+metals system	70
nonlinear_system_solver/constrained_equilibrium_probe.f90	standalone re-evaluation of one saved constrained-equilibrium state, without the hydrodynamics	15
simulation-build new-module total		36 696

The other 106 are the standalone test drivers under `src/tests/`, which did not exist in ATES at all. They state what a routine is allowed to return and are run by suite, not by the production binary:

Test driver	What it asserts	SLOC
tests/krylov_and_dogleg/krylov_and_dogleg_tests.f90	what the linear solve and the step geometry of the stationary solver are allowed to return	2751
tests/steady_species_rows/steady_species_rows_tests.f90	the unknown space of the stationary system, one row per transported balance	1826
tests/ionization_stage_flux/ionization_stage_flux_tests.f90	the stage fluxes sum to the element nucleus flux at every face, against the formula of the derivation and not against the module itself	1536
tests/carrier_retry/carrier_retry.f90	the carrier-local checkpoint and the rejection-and-retry controller, on a real molecular column	1463
tests/diffusion_tests.f90	the binary H/He element-diffusion operator: analytic limits and conservation	1055
tests/certification/certification_contexts.f90	the certification evaluator and its contexts, against the predicates they replace	1050

Test driver	What it asserts	SLOC
tests/grid_and_gates/hydrostatic_residual.f90	the momentum row returns zero on a hydrostatic state	1031
tests/carrier_retry/closure_returned_state.f90	what the thermochemical closure hands back on a molecular column: the composition, its chemical residual, the conditioning, and the state a refused trial puts back	890
tests/stage_row_balance/stage_row_balance_tests.f90	the signed sum of the gross channels of an ion-stage row is that stage's source, in molecular and in atomic gas and with the He 2 ³ S level tracked and untracked	874
tests/element_operator/element_operator_tests.f90	the boundary the element transport operator is measured under	852
tests/physics_probe/molecular_energy_recipients.f90	the enthalpy of a reaction is untouched by where its energy goes: heat, radiation and internal energy close on the one formation table, and the R12/R15 pair stays in detailed balance collider by collider	750
tests/grid_and_gates/base_boundary_continuity_probe.f90	the base face density does not move along a perturbation of the wind window once the branch is decided, and the two limits coincide at a vanishing window	592
tests/h2_level_ladder/h2_ladder_model.f90	a reduced statistical equilibrium of the H ₂ ground-state ladder from the published level, radiative and collisional data, which brackets the scalar thermalized fraction	561
tests/species_face_flux/species_face_flux_tests.f90	species advection on the face mass fluxes, and the element rows of the diffusion operator	549
tests/low_mach_stress_energy/low_mach_stress_energy_form.f90	the discrete kinetic-energy quadratic form of the gated fourth-difference stress, and the sign a coefficient carried outside both differences does not have	519
tests/physics_probe/weno3_reconstruction_order.f90	convergence order of the face reconstruction against the exact face value	482
tests/spectrum_type/loaded_sed_table_semantics.f90	what a row of a loaded SED table means, and what the production consumers make of it	417
tests/attempted_step/attempted_step_tests.f90	the checkpoint and the step-size policy of the attempted-step controller	415
tests/a2_m1/a2_m1_rate_check.f90	the oxygen rate coefficients against their published sources	406
tests/charge_exchange_rows/transported_stage_sources.f90	the charge-exchange source of a transported ionization stage is the source the local systems' rows carry, in each of the three bases the metal systems assemble	399
tests/charge_exchange_rows/reaction_source_orientation.f90	one electron transfer moves $-R$, $+R$, $-R$, $+R$ over four stages, creates no nucleus and makes no net charge, in each of the three equation bases	395
tests/spectrum_type/balmer_band_quadrature.f90	quadrature of the Balmer-continuum band integrals of excited_hydrogen	367
tests/energy_update/conduction_floor_tests.f90	what the Crank–Nicolson transport stage returns, refuses and conserves	360
tests/coupled_block_jacobian/coupled_block_linear_system.f90	the directional action of the coupled block on one frozen iterate: its repeatability, and its perturbation ladder row group by row group	353
tests/physics_probe/metal_photoionization_fits.f90	every metal photoionization cross section against the published fit, and only where it is valid	343
tests/element_census_tests.f90	the elemental and charge invariants of a composition state	340

Test driver	What it asserts	SLOC
tests/energy_update/energy_update_ tests.f90	the residual-controlled temperature solve of the implicit energy source step	334
tests/grid_and_gates/base_closure_ candidates_probe.f90	two discriminants for the same characteristic base condition, read on one state under the production stationary operator	328
tests/carrier_helium_inventory/carrier_ helium_inventory.f90	the helium a cell holds is the helium it had: the admissible stage partition leaves out the nucleus inside HeH^+ and the population in $\text{He } 2^3\text{S}$	310
tests/spectrum_type/loaded_sed_threshold_ edges.f90	the photon grid built from a loaded SED, and the rates formed on it	304
tests/boundary_state/contact_law_at_the_ base.f90	the lower boundary as a contact: a column on the reservoir isentrope stays at rest, a pressure mismatch launches the acoustic response, and the entropy is upwinded on the matched face velocity	292
tests/ionization_imposed_fractions/ imposed_ionization_rows.f90	an imposed ionization row is the identity $x - x_{\text{fix}}$ and is one number in all seven systems, and nothing depends on the carried values where nothing is imposed	288
tests/coupled_block_handover/coupled_ block_handover_contract.f90	the transition into the coupled carrier solve is reached once and from the recorded entry state bit for bit, never from a live token the stalling pass cannot carry	285
tests/h2_level_ladder/h2_ladder_ statistical_equilibrium.f90	the solved ladder against the first-event branching fraction: detailed balance and the thermal limit, and the four ways an injected formation energy leaves	275
tests/physics_probe/species_formation_ energy_table.f90	the formation-energy table of the energy ledger, and the reaction energies that are its differences	275
tests/spectrum_type/wasp121b_sed_ sensitivity.f90	the production band quantities of the three WASP-121 b spectrum candidates	273
tests/physics_probe/lya_band_ transmission.f90	the stellar $\text{Ly}\alpha$ band reaching a cell through the atomic hydrogen column above it	272
tests/coupled_source_step/coupled_source_ tests.f90	the coupled temperature-composition source step	260
tests/charge_exchange_rows/he_h_pair_ reservoir.f90	the $\text{He} \leftrightarrow \text{H}$ charge-exchange pair reacts from the ground singlet, the same one in every system carrying helium and a metastable	253
tests/physics_probe/lyman_werner_cell_ mean.f90	the Lyman–Werner rate a cell carries is the mean of the local rate over that cell	234
tests/charge_exchange_rows/metal_helium_ reservoir.f90	the metal + He reactions of group C react from the ground singlet, the same one in the three systems that assemble them	231
tests/physics_probe/lya_beam_cell_mean.f90	the stellar $\text{Ly}\alpha$ field of a cell is the beam mean over the cell’s own optical depth	229
tests/spectrum_type/balmer_continuum_ field.f90	the two scalar Balmer-continuum rates of $\text{H}(n=2)$	227
tests/carrier_returned_state_acceptance/ carrier_returned_state_acceptance.f90	what makes a returned carrier state a solution, and what does not	225
tests/physics_probe/positivity_limiter_ scaling.f90	the positivity scaling of the reconstructed face states (Zhang & Shu 2010)	223
tests/grid_and_gates/photon_grid_ threshold_edges.f90	every active absorber’s threshold against the bin partition of the photon grid	213
tests/physics_probe/photoionization_field_ substitution.f90	the order in which the field and the composition that attenuates it are brought into agreement	209

Test driver	What it asserts	SLOC
tests/adv_static_limit/stationarity_by_the_mass_row.f90	which test decides that the state the post-process was handed is stationary in a cell	199
tests/carrier_outer_boundary/carrier_outer_boundary.f90	the carrier column is closed to diffusion at both ends, one outflow ghost rule serves every evaluation, and the advective band entries are the operator's own derivative	196
tests/physics_probe/co_destruction.f90	the CO destruction rates, the shielding table, the deposited energies and the cell mean	195
tests/carrier_reference_scales/carrier_reference_scales.f90	the element a transported carrier is made of, and whether its chemistry rows can be differentiated	191
tests/physics_probe/recombination_coefficient_fits.f90	every recombination coefficient against the published fit	189
tests/spectrum_type/photon_grid_n2_floor.f90	the photon grid built when the excited-hydrogen coupling is armed	184
tests/charge_exchange_rows/advection_pair_reservoir.f90	the advection-corrected system hands the He $\leftrightarrow$ H pair the same ground singlet the equilibrium system does	182
tests/carrier_constraint_attribution/carrier_constraint_attribution.f90	which residuals belong to the solver and which to a constraint	179
tests/adv_static_limit/enthalpy_flux_of_mass_divergence.f90	the third term of the advected energy equation	178
tests/grid_and_gates/base_branch_discriminant.f90	what the base contact reads its direction off: the wind window wherever it carries one flux, the matched face velocity elsewhere, never the first cell's own product	174
tests/grid_and_gates/free_outflow_boundary.f90	the outer free-outflow boundary on a supersonic wind, and its ghost state	174
tests/spectrum_type/planck_photon_field.f90	the incident flux put on the photon grid for the photospheric blackbody type	173
tests/carrier_boundary_jacobian/carrier_boundary_jacobian.f90	the boundary derivatives of the carrier advection, a copied ghost carrying one and a stated one carrying none	165
tests/a2_m2/a2_m2_kinetics_check.f90	the kinetics quantities the oxygen option adds on top of the rate audit	163
tests/physics_probe/lya_escape_probability.f90	the escape probability of the two-level closure against the published slab solution	163
tests/e1_h2/e1_h2_channel_check.f90	the mutually exclusive H ₂ photoabsorption channels	161
tests/adv_static_limit/upstream_caloric_energy.f90	the upstream energy of the advected energy equation	154
tests/physics_probe/roe_low_mach_velocity_jump.f90	the low-Mach scaling of the normal velocity jump on the ROE branch, against the analytic three-wave flux (Rieper 2011)	152
tests/physics_probe/photoionization_field_self_consistency.f90	the field one cell is solved in, against the composition that cell returns	148
tests/physics_probe/co_helium_ion_sink.f90	He ⁺ + CO as a species sink, not only as a heat source	147
tests/physics_probe/h3p_cooling_limits.f90	the limits, the continuities and the published values of the H ₃ ⁺ cooling rate	147
tests/grid_and_gates/photon_grid_quadrature.f90	the photoionization rates of one unattenuated cell against the band quadrature	146
tests/physics_probe/riemann_wave_speeds.f90	signal speeds at a face, and the state one conservative update with that flux leaves behind	143
tests/physics_probe/molecular_third_body_in_the_chemistry.f90	the three-body collider sum of the R12/R15 pair is one number for the composition and for the energy, and is not the total density	140

Test driver	What it asserts	SLOC
tests/physics_probe/helium_level_cooling_ledger.f90	which helium level each cooling channel of eval_cool is charged to	139
tests/physics_probe/heating_channel_closure.f90	one assembly of the volumetric heating rate, and the closure of its channels	138
tests/acceptance_classes/acceptance_class_contract.f90	one meaning of the ionization acceptance classes for every consumer	132
tests/physics_probe/opacity_model_p_ledger.f90	the radiative ledger of a cell whose cross sections are broadened by opacity model P	124
tests/physics_probe/photoevent_energy_ledger.f90	the energy an isolated photoabsorption event has to account for	117
tests/physics_probe/xuv_cell_mean_attenuation.f90	the beam attenuation the photoionization rate of a cell is evaluated with	115
tests/grid_and_gates/grid_window_indices.f90	the two index statements that decide which cells a quantity is read from	101
tests/grid_and_gates/base_cell_width_provenance.f90	the base cell width is written as a decimal that reads back to the same bits, and the default and the pinned line state the widths their runs were made on	98
tests/physics_probe/h2_rovibrational_identity.f90	the identities tying the H ₂ partition function of the chemistry to the internal energy of the caloric EOS	96
tests/adv_static_limit/stationary_flux_refusal.f90	the sensitivity of the advected energy correction to a non-stationary mass flux	91
tests/physics_probe/ionization_threshold_turn_on.f90	one definition of each ionization threshold, and every cross section turning on exactly there	90
tests/grid_and_gates/base_grid_pinned_width.f90	which grid each of the four base cell widths builds, compared on the bits of every cell center	86
tests/grid_and_gates/grid_width_identity.f90	the cell-width and total-width identities of the radial grid	83
tests/physics_probe/h2_channel_detector_ratio.f90	the measured proton-to-H ₂ ⁺ ratio of Chung et al. (1993) against the four channels	83
tests/physics_probe/fine_structure_escape_probability.f90	the escape probability of the ground-term fine-structure lines, and the depth it is fed	80
tests/physics_probe/atomic_mass_and_radius_constants.f90	the helium mass and the astronomical constants of the input readers, one definition each	79
tests/physics_probe/assertion_report.f90	one printed verdict line per assertion, and the failure count the driver exits on	74
tests/adv_static_limit/thermal_damkohler_limit.f90	the validity condition of the advection-corrected energy solve, as a function of the flow speed	72
tests/a2_m2/a2_m2_g3_run_check.f90	with every photolysis rate at zero, the solved O/OH/H ₂ O partition is the chemical equilibrium	71
tests/physics_probe/absorbed_fraction_switch.f90	$(1 - e^{-d})/d$, the fraction of the photons entering a cell that the cell absorbs	71
tests/constrained_network_layout/transported_proton_constraint_layout.f90	which species are unknowns of the continuation, and whether the system built is square	71
tests/test_columns.f90	synthetic columns whose species carry the mass density they are measured against	67

Test driver	What it asserts	SLOC
tests/physics_probe/h2_infrared_line_populations.f90	the LTE populations weighting the H ₂ quadrupole lines are those of the caloric EOS ladder	65
tests/charge_exchange_rows/turnover_cell_rates.f90	the charge-exchange turnover bound is the loaded cell's, carries no history, and is each thread's own	62
tests/physics_probe/wind_ae_bridge_composition.f90	the composition an EXHALE input.inp hands the Wind-AE solver	54
tests/spectrum_type/fuv_band_quadrature.f90	the band integral of a loaded spectrum against closed-form answers, with the band edges inside segments, on nodes and at the table ends	53
tests/physics_probe/voronov_carbon_ionization.f90	the electron-impact ionization of C ⁰ and C ⁺ against the published Voronov (1997) rows	51
tests/physics_probe/water_photolysis_lyman_alpha_yields.f90	the H ₂ O branching on the band that contains Ly α , against the published yields	49
tests/physics_probe/h3p_recombination_branching.f90	the two product channels of H ₃ ⁺ dissociative recombination	44
tests/energy_update/cooling_models.f90	analytic cooling laws for the tests of the implicit energy update	43
tests/species_masses/carrier_background_masses.f90	the mass the carrier transport gives each of its background collision partners	36
tests/charge_exchange_rows/stale_cell_rates_refused.f90	a residual assembled at a temperature the rate coefficients were not filled at is refused	30
tests/certification/certification_capacity.f90	one equation past a full certification report is refused by name, before any field is written	26
tests/charge_exchange_rows/stale_cell_turnover_refused.f90	a turnover bound taken at a temperature the rate coefficients were not filled at is refused	26
tests/charge_exchange_rows/stale_cell_stage_sources_refused.f90	a stage source evaluated at a temperature the rate coefficients were not filled at is refused	23
test-driver total		32 799
new-module total		69 495

So the 162 new core modules contribute 69 495 of the 91 633 added code lines; the remaining 22 138 added (and 1 150 of the 1 219 removed, the other 69 being the deleted HLLC estimator) are edits spread across the 38 modified files. The ten heaviest, by added code lines, are the main driver (EXHALE_main.f90, 4 396 added, 246 → 4 511 SLOC: the marching loop, the attempted-step adoption boundary, the run-mode contracts and the written-state assertions), the cooling module (Cool_coeff.f90, 3 560 added, 228 → 3 691 SLOC: CHIANTI metal-line fits, Fe tables, fine-structure statistical equilibrium, Penning and molecular rate coefficients), the ionization equilibrium and its diagnostics (ionization_equilibrium.f90 2 767 and util_ion_eq.f90 1 813: the metal, molecular and oxygen networks, the acceptance classes, the photoionization field and the heating channels), input parsing (input_read.f90, 1 871), the state reader and writer (load_IC.f90 1 529, write_output.f90 989, write_setup_report.f90 858), the advection-corrected post-processor (post_process_adv.f90, 665) and the cross sections (cross_sec.f90, 438). Those ten carry 18 886 of the 22 138. The rest of the edits are in the reconstruction, the fluxes, the boundary conditions, the grid, the time step, the energy equation and the System_* ionization systems, which wire the metals, the molecules, the opacity dispatcher, the Roche potential, the transonic/auto/Wind-AE IC, the excited-H and Ly α physics, the element diffusion and the Brent/Newton solver upgrades into the original solver.

In total the EXHALE core is now $\approx 26.7\times$ the original code base (3 514 $\rightarrow$ 93 928 SLOC), or $\approx 17.4\times$ counting only the simulation build (61 129 SLOC), touching almost all of the original files (38 of 47 modified, one deleted, plus 162 new); with the ported Wind-AE IC generator the full Fortran tree is 244 files / 98 684 SLOC ($\approx 28.1\times$). Beyond Fortran, the tree also carries the Python tooling (GUI, the EXHALE_transit.py transmission post-processor, plotting/analysis, the lower-atmosphere adapters and the elemental-flux closure driver) and 304 946 lines of documentation (390 md/tex files, notebook checkpoints excluded: the memos, the manual, the dated audit directories and the update logs) under docs/.

Fortran source inherited unchanged from ATES

Of the 47 upstream Fortran files (ATES/ATES-Code-main), **8 are byte-for-byte identical** in EXHALE (re-measured 2026-09-22, and the same eight as on 2026-09-11: nothing entered or left the set in between): the numerics EXHALE reuses without touching, which are now the linear-algebra part of the included MINPACK nonlinear solver plus the gravity pre-evaluation. Table 11 lists them with their role. No file is identical after whitespace and case normalization but not byte-identical, so the normalized-unchanged set and the byte-identical set are the same eight.

Five files left the byte-identical set since the 2026-08-30 edition of this table, each for a documented reason rather than drift.

- nonlinear_system_solver/hybrd.f90, hybrd1.f90 and fdjac1.f90 carry a new unit_step_floor argument threaded from hybrd to fdjac1, which selects the forward-difference step of the Jacobian: `.false.` is MINPACK's own $h = \varepsilon|x_j|$ with the exact-zero fallback, `.true.` floors it at unit magnitude, $h = \varepsilon \max(1, |x_j|)$, which is the rule for unknowns already scaled so that unity is their natural size (the logarithmic unknowns of the constrained chemical equilibrium). The params dummy also became assumed-size, so one transport array serves systems of different length. hybrd1 passes `.false.` and is therefore unchanged in behavior.
- states/Source.f90 gained the well-balanced option of 2026-09-10 (item N37 of the stage-2 solver program): under `Well balanced: True` the gravitational source and the geometric pressure term are not evaluated here at all, because they cancel the equilibrium part of the momentum flux difference identically and RK_rhs assembles the row from what is left (Käppeli & Mishra 2014, J. Comput. Phys. 259, 199, eq. 2.26). The file also reads its pressure through `caloric_eos` instead of a constant γ .
- radiation/J_inc.f90 carries the single-spectrum-type decision: `Spectrum type:` now states the spectrum of the whole photon grid, the XUV and the part below 13.6 eV alike, and no band is filled from a type the input did not select. The file also became the one definition of the eV-to-Hz conversion and of the Balmer-continuum head $e_{\text{th,HI}}/4$.

The two wave-speed estimators are no longer in the “effectively untouched” category either. `flux/speed_estimate_HLLC.f90` was deleted as dead code, as recorded above. `flux/speed_estimate_R0E.f90` was rewritten (99 code lines added, 43 removed) into a star-state estimate that returns a status: the primitive-variable, two-rarefaction and two-shock branches of Toro (2009) chapter 9 are selected by the pressure ratio, the vacuum condition of his eq. 4.76 is tested first, and a face with no admissible star state is reported rather than given a square root of a negative ratio.

Table 11: The 8 Fortran files byte-identical between the original ATES (ATES-Code-main) and EXHALE (measured 2026-09-22), grouped by role.

File	Purpose
<i>MINPACK linear algebra and constants</i> (Argonne; dir nonlinear_system_solver/)	
qrfac.f90	QR factorization via Householder transformations
qform.f90	Accumulate the orthogonal factor Q from the QR factorization
r1updt.f90	Rank-one update of the QR factorization
r1mpyq.f90	Apply the Givens-rotation sequence of the rank-one update
dogleg.f90	Dogleg (trust-region) step combining Gauss–Newton and steepest descent
enorm.f90	Euclidean norm of a vector (overflow/underflow safe)
dpmpar.f90	Machine floating-point precision / range constants
<i>Grid setup</i>	
init/set_gravity_grid.f90	Pre-evaluate gravity at cell centers and interfaces